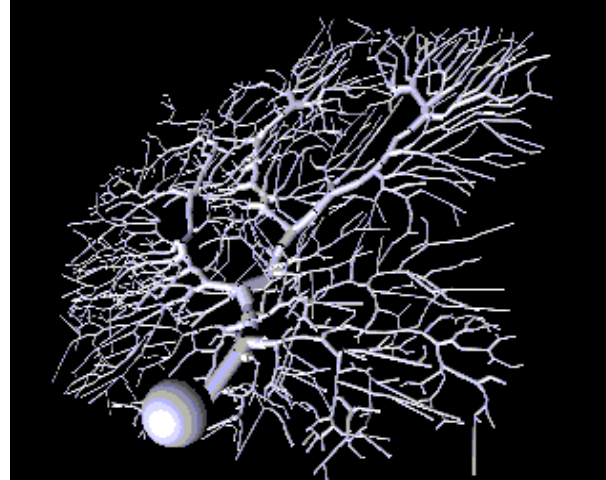
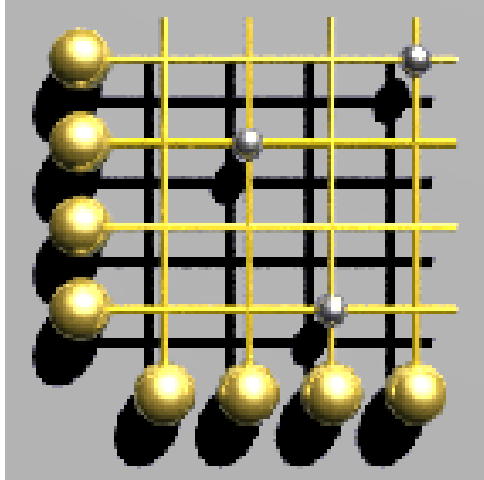


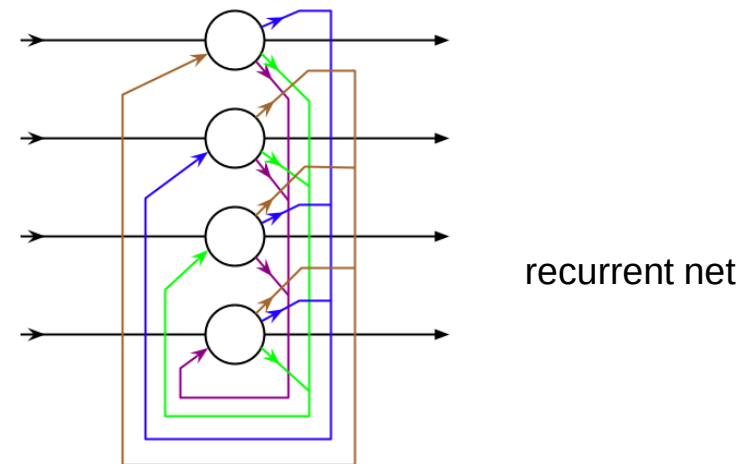
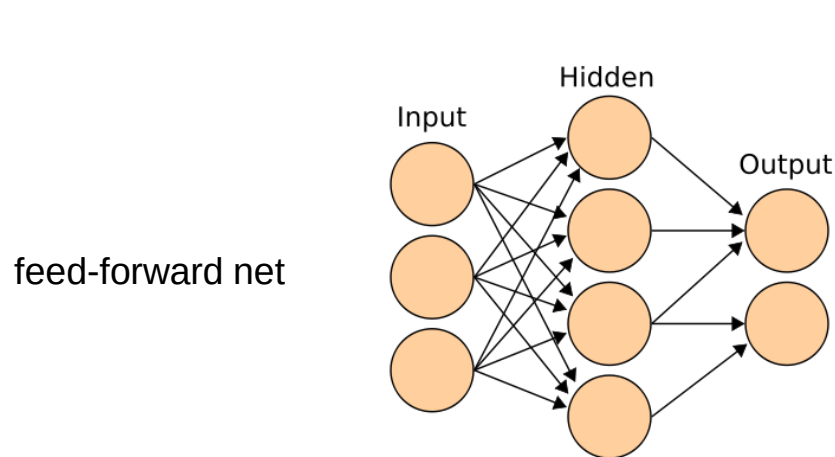
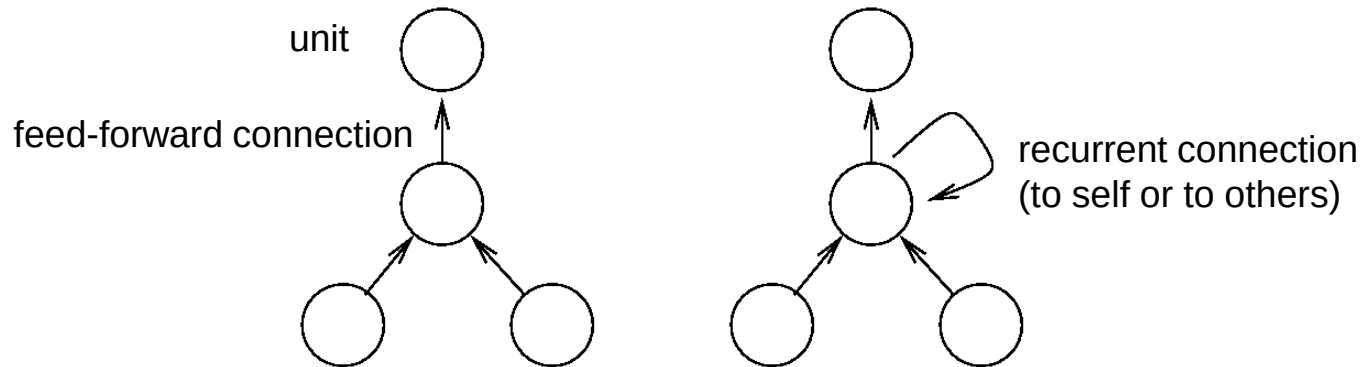
Neural Networks and Machine Learning

Introduction to neuroscience and neural networks



What are neural networks?

Networks of simple computational units that share some properties with real neurons.

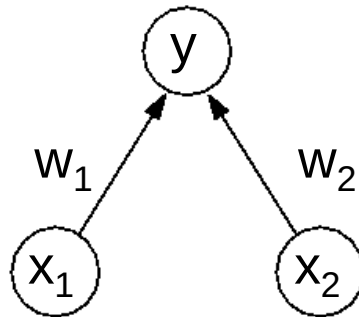


Neural networks can be feed-forward or recurrent.

Basic properties of neural networks

The **activation** of a unit depends on the activation of the units it receives inputs from and the **weights** of the incoming connections.

output unit with activation $y = f(x_1, w_1, x_2, w_2)$



connections with
weights w_1 and w_2

two input units with activations x_1 and x_2

Why study neural networks?

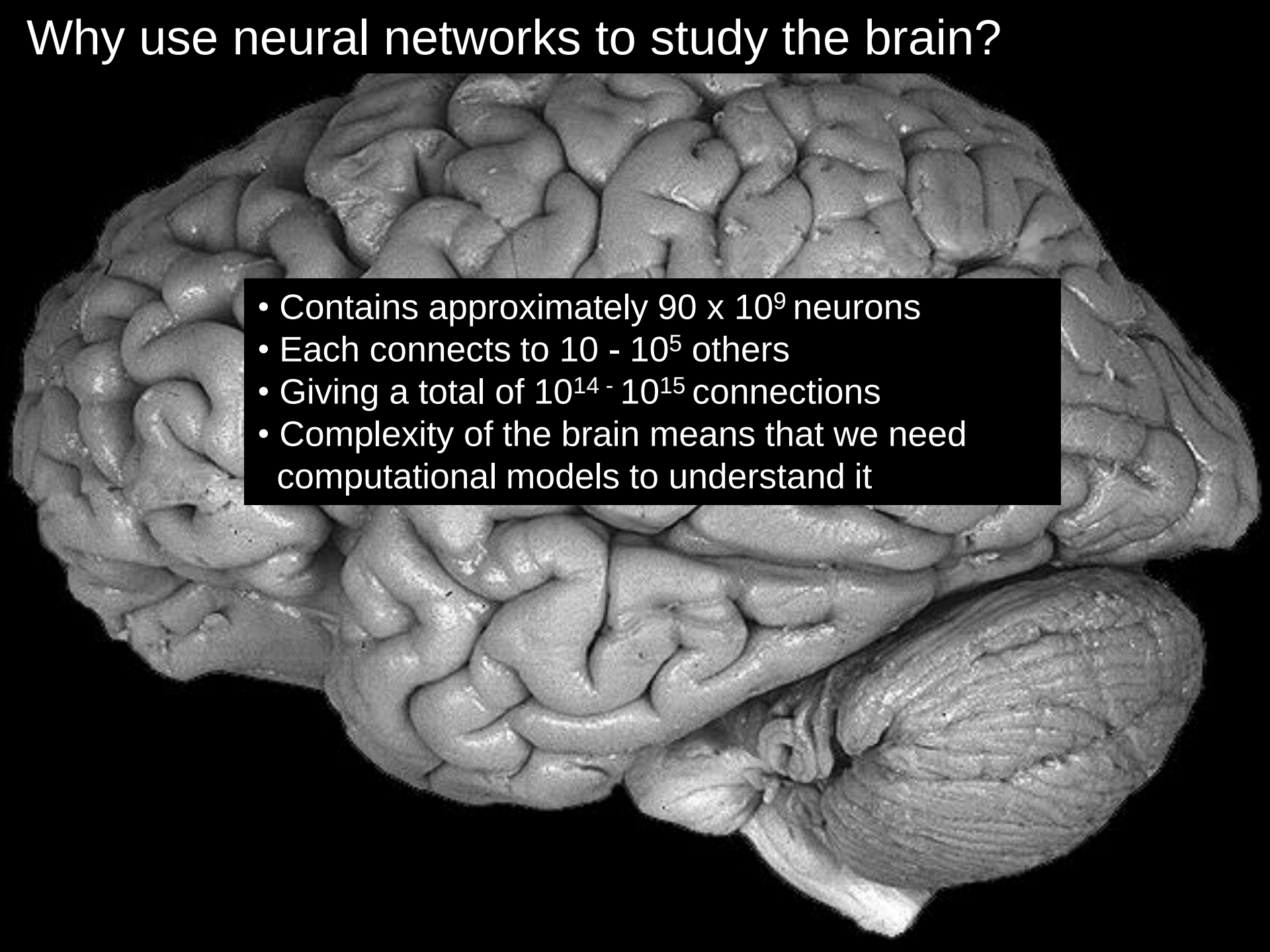
Understand basic principles of neural computation

- Curiosity: how does the brain work?
- Give insight in diseases such as epilepsy, Alzheimers, schizophrenia, Parkinsons, etc.

Use insights from neuroscience to solve engineering tasks

- Character recognition
- Speech recognition
- Learning to pronounce English text
- Medical diagnosis of heart attacks
- Control of autonomous vehicles
- Recognition of facial expressions
- Predicting skin penetration of drugs
- Improving optical communication systems
- etc.

Why use neural networks to study the brain?

- 
- Contains approximately 90×10^9 neurons
 - Each connects to $10 - 10^5$ others
 - Giving a total of $10^{14} - 10^{15}$ connections
 - Complexity of the brain means that we need computational models to understand it

Structure of a neuron

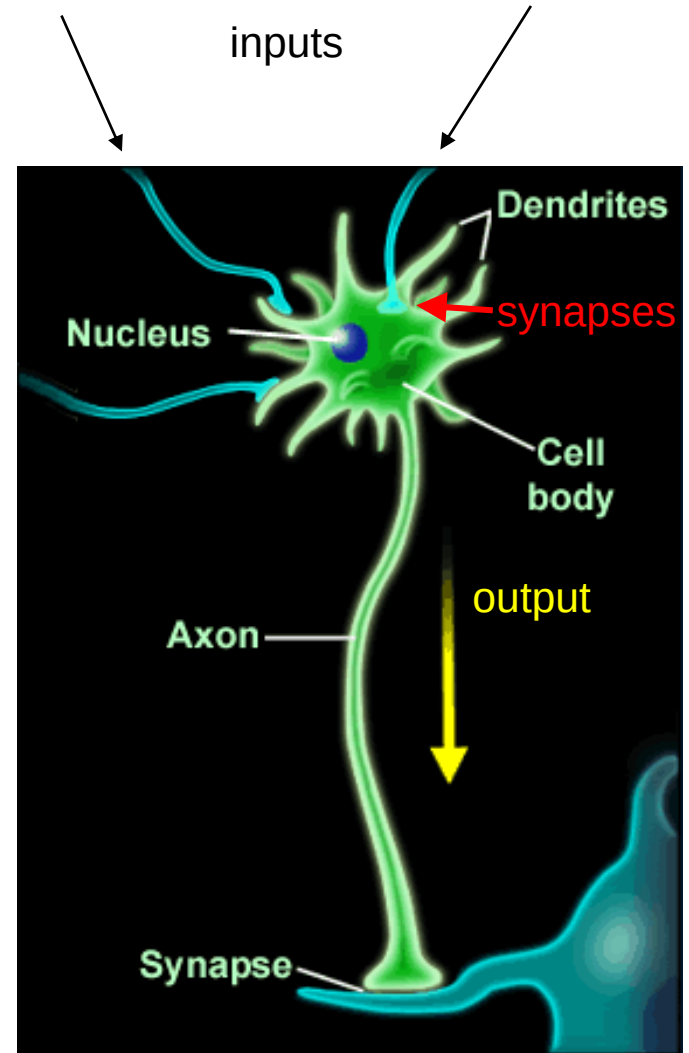
A typical neuron consists of three main parts:

Dendrites: receive and process inputs.

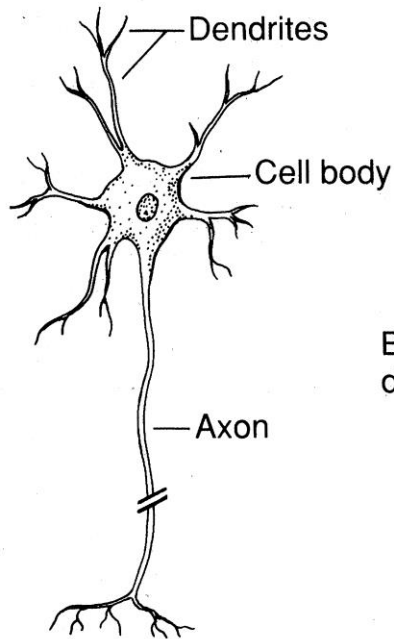
Soma (cell body): integrates inputs.

Axon: generates output (at the axon hillock or initial segment) and transmits it to target neurons.

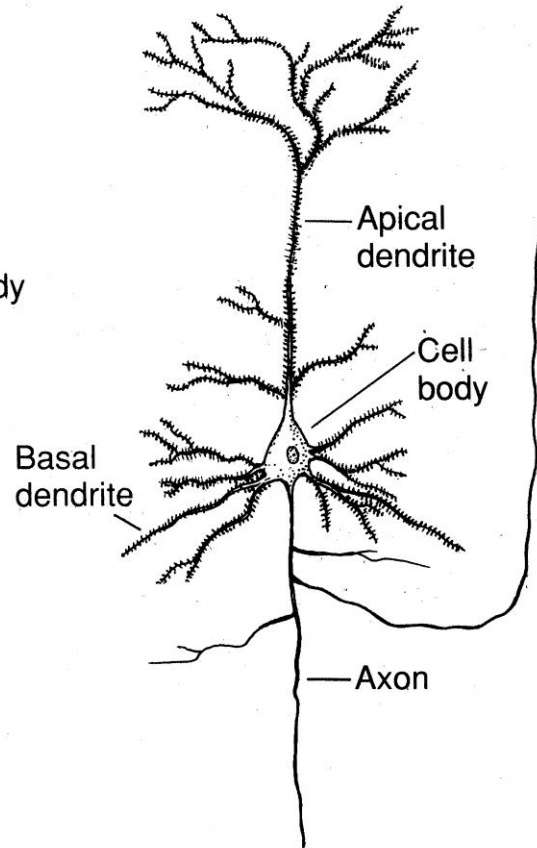
The connection from an axon of one neuron to a dendrite of another neuron is called a **synapse**.



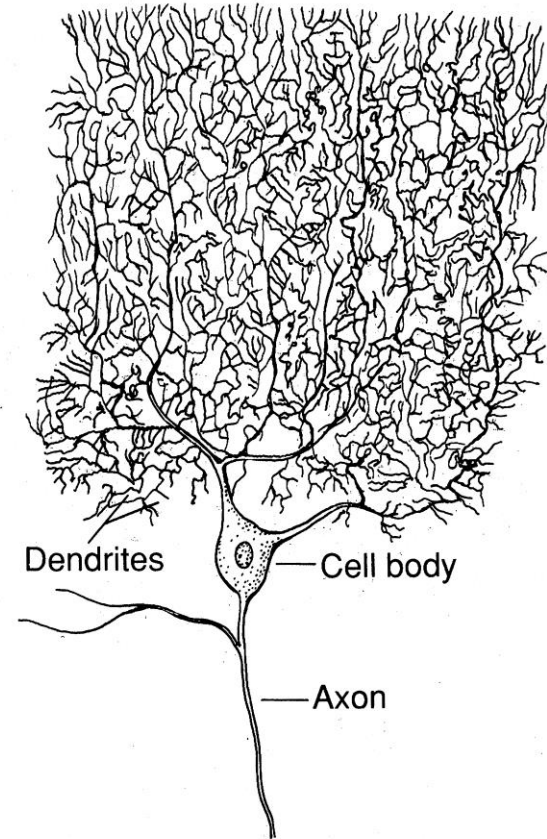
Variability of neuronal structures



Spinal motor neuron



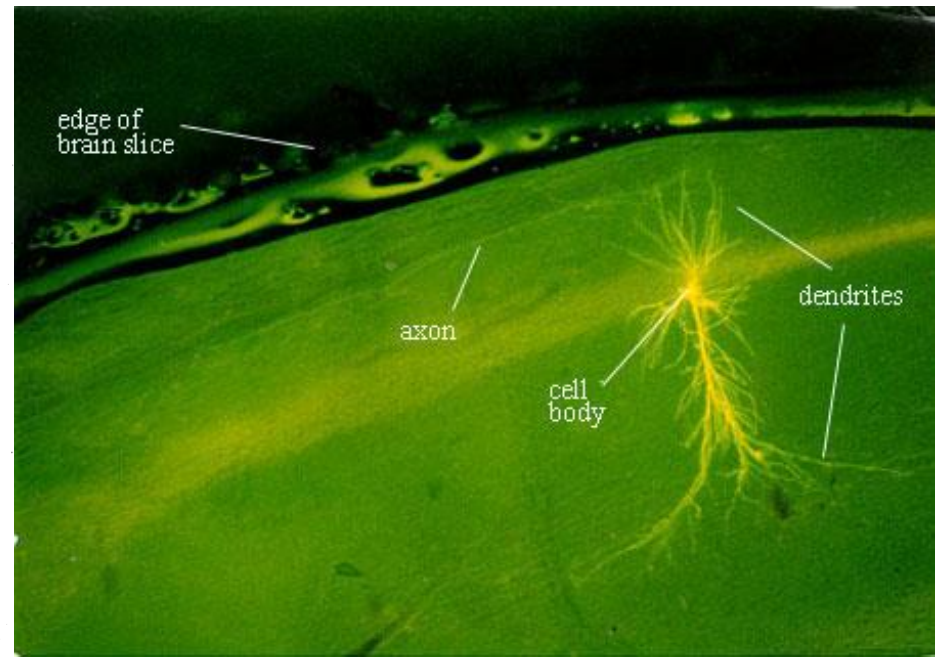
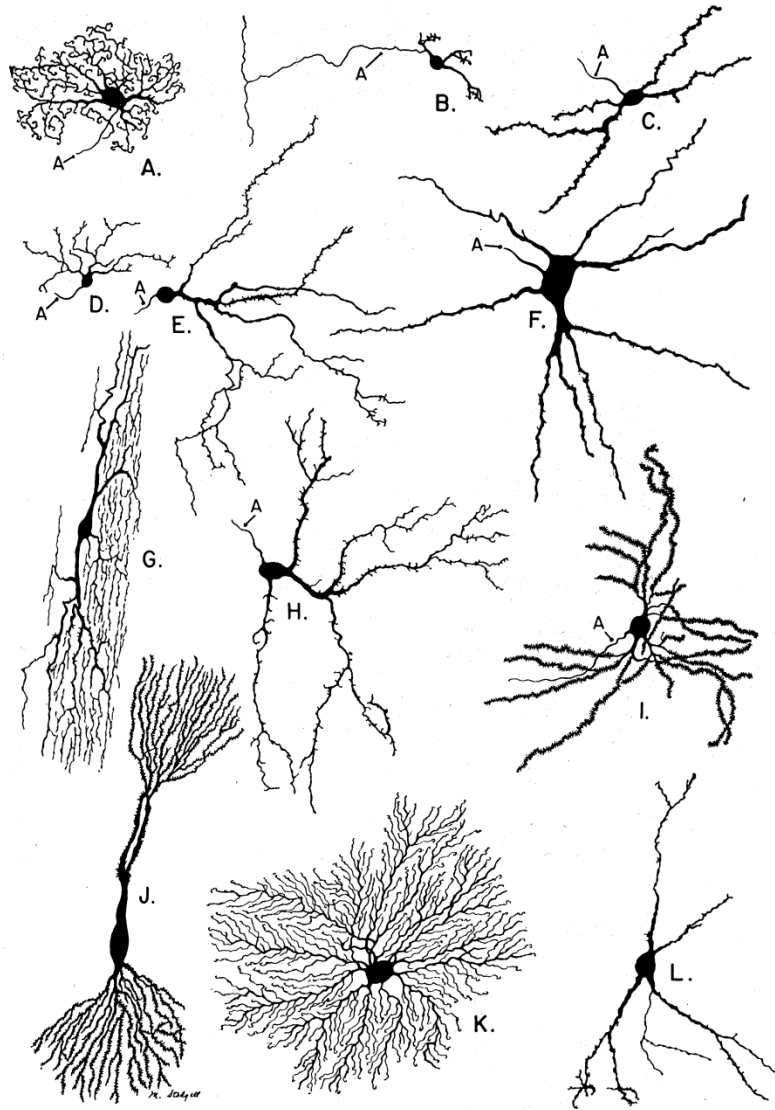
Hippocampal pyramidal cell



Purkinje cell of cerebellum

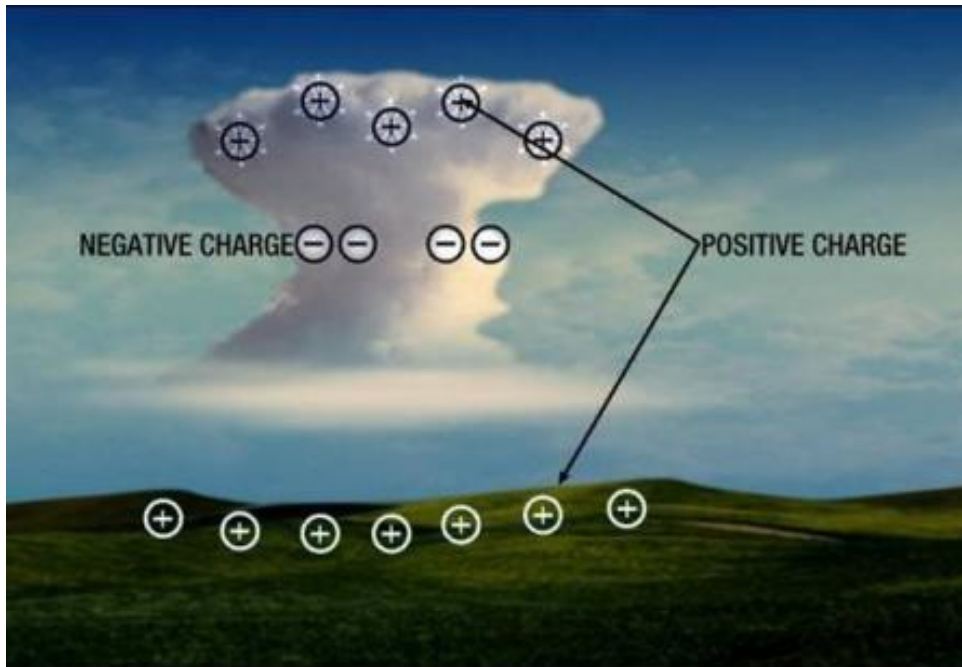
In addition to synapses from axons onto dendrites, there are also synapses from axons onto cell bodies (somata) and onto other axons.

More neuronal structures



The functional implications of these different structures are still not clear!

How do neurons work?

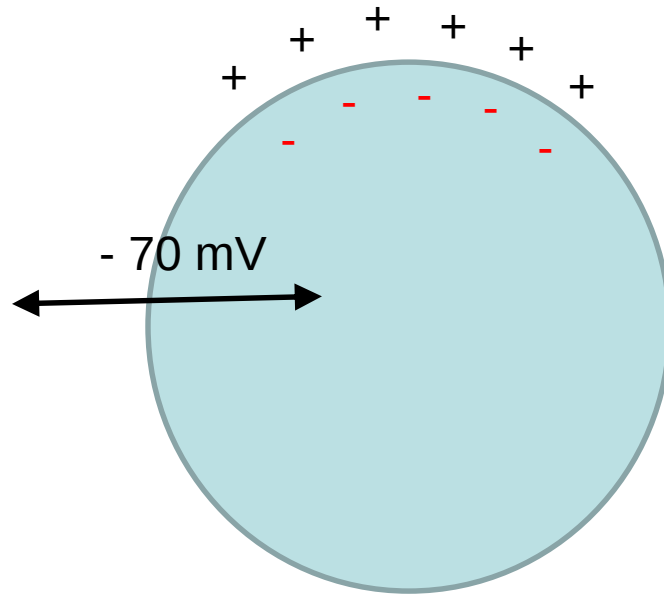


Neurons operate based on the principle of charge separation. The separation of positive and negative charges results in the generation of voltage gradients and electrical currents (as in thunderstorms).

Charge separation by neurons

Neurons can separate charges as they have a cell membrane that separates the inside from the outside.

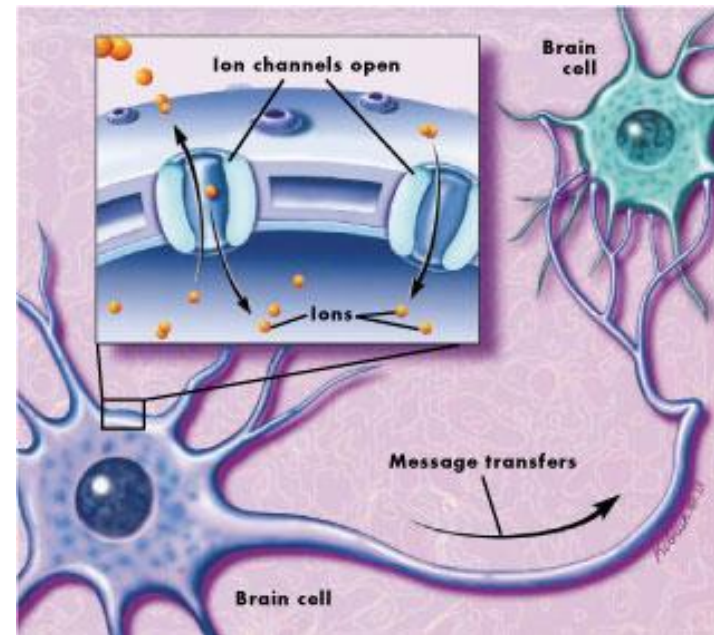
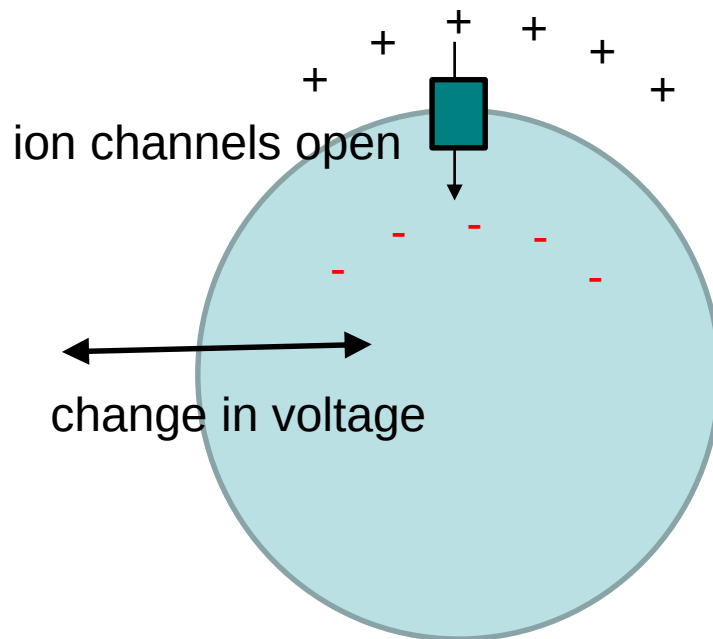
Most of the time (in the absence of any input), the inside is negative relative to the outside and the resting membrane potential or voltage is around -70 millivolts (mV).



Neurons are electrical devices

The transmission of information by neurons is based on temporary changes in their membrane potential (voltage).

These changes in membrane voltage are based on the opening and closing of ion channels (small pores in the neuronal membrane) that result in the movement of charges (ions) across the membrane.

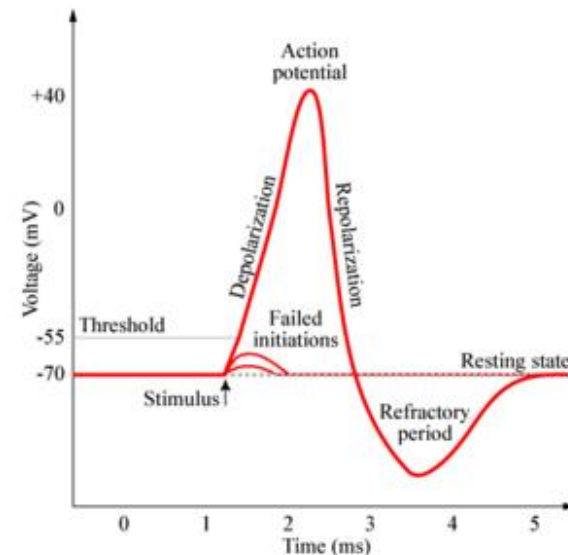
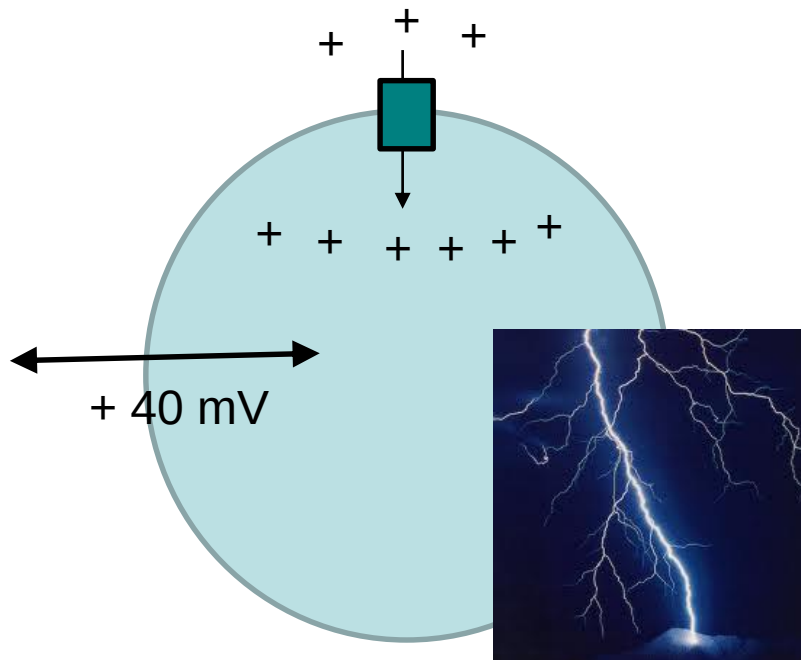


Neurons signal by generating action potentials

In the absence of any input: inside is negative relative to outside, resting membrane potential around -70 mV.

Sufficiently strong input to the neuron → ion channels open → movement of positive charges (sodium ions) into the neuron → inside becomes more positive than outside → a positive membrane potential ($\sim +40$ mV) called action potential (AP) is generated [\[demo\]](#)

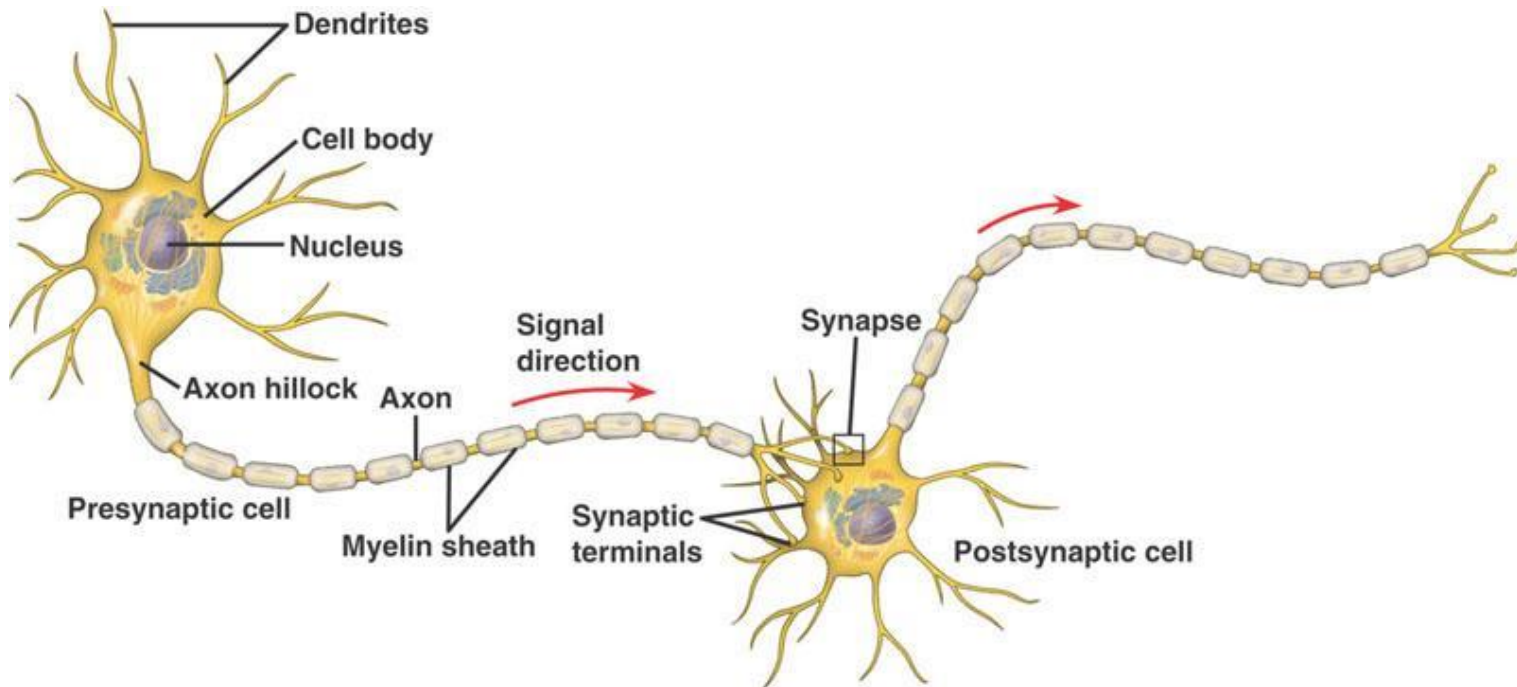
APs are all-or-none: generated if and only if a firing threshold is reached.



Action potential propagation

The initial generation of an AP usually takes place in the axon hillock (segment of the axon just next to the soma).

The AP is then propagated along the axon to the locations where the axon forms synapses with other neurons.



[\[demo from Matthews: Neurobiology\]](#)

Chemical transmission across synapses

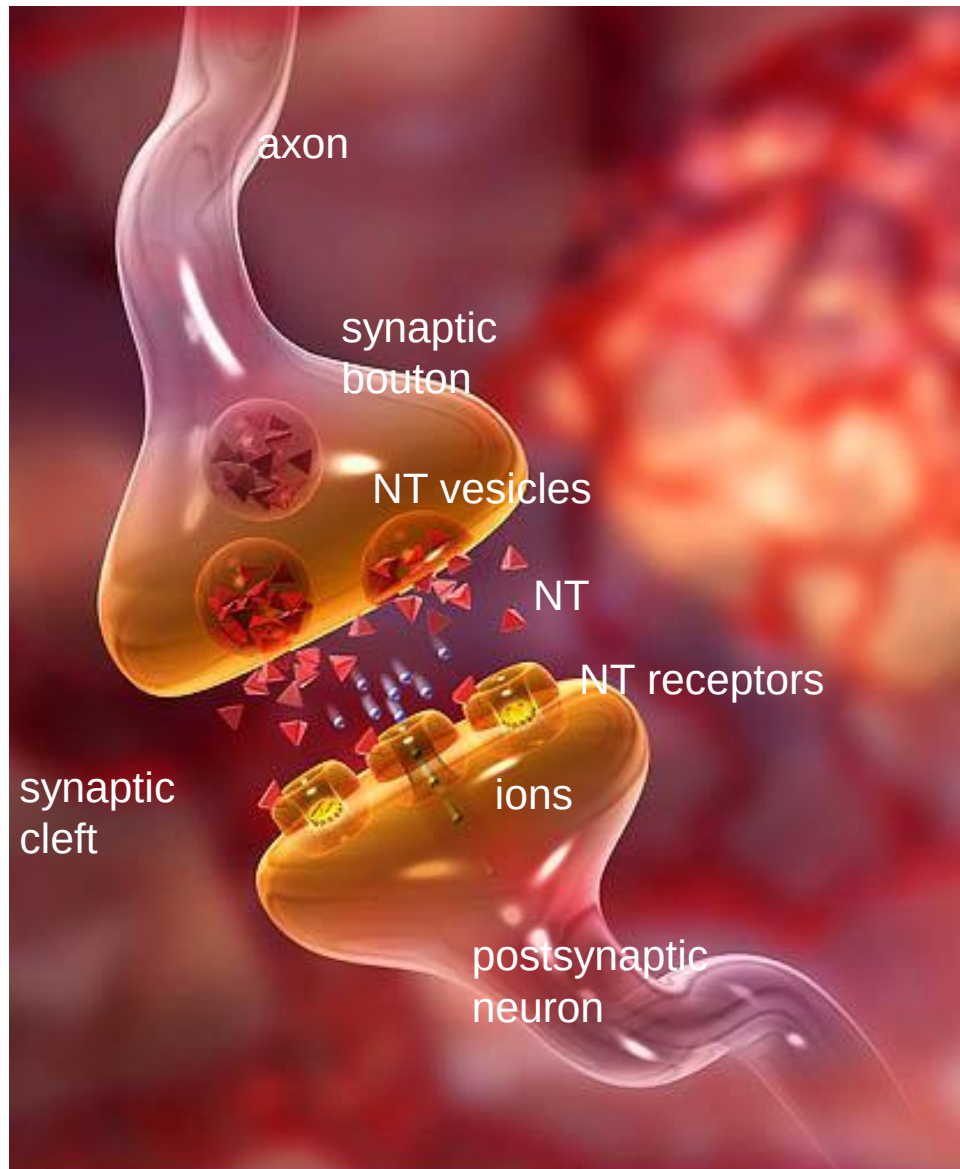
The AP is propagated down the axon to synaptic boutons that make contact with postsynaptic neurons.

When the AP arrives at the synapse, a neurotransmitter substance (NT) is released from stored vesicles.

The NT diffuses across the synaptic cleft and binds to receptor proteins on the postsynaptic membrane.

Many NT receptors are ion channels. Binding of NT opens them and results in an inward flow of positively charged ions. This generates a postsynaptic potential (PSP) in the postsynaptic neuron.

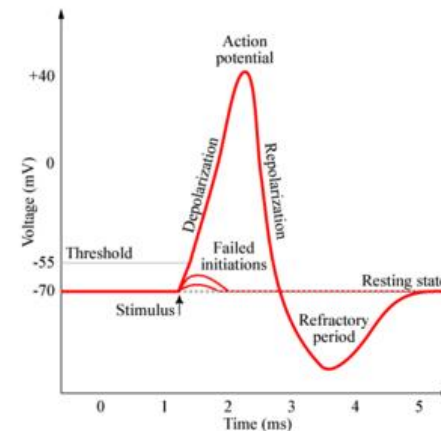
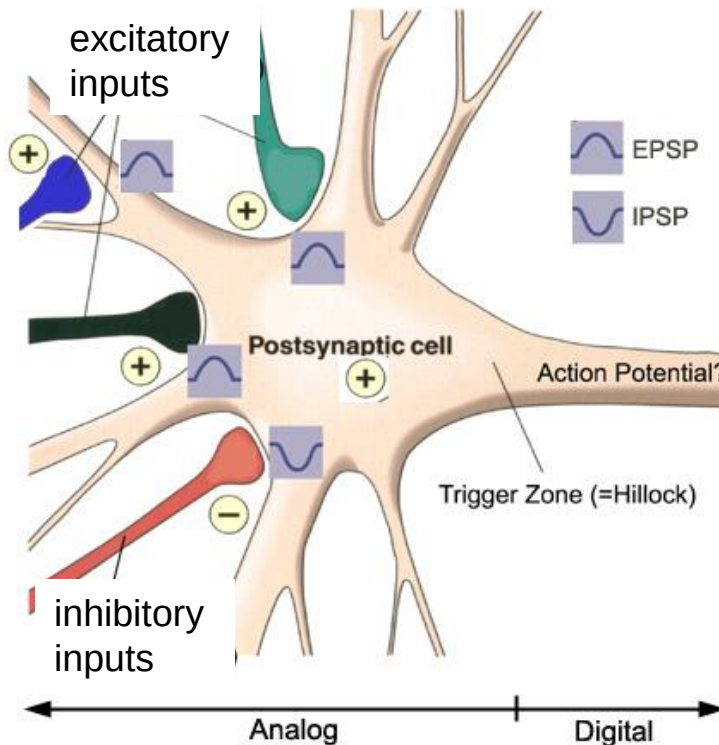
[\[demo from Matthews: Neurobiology\]](#)



Postsynaptic potentials

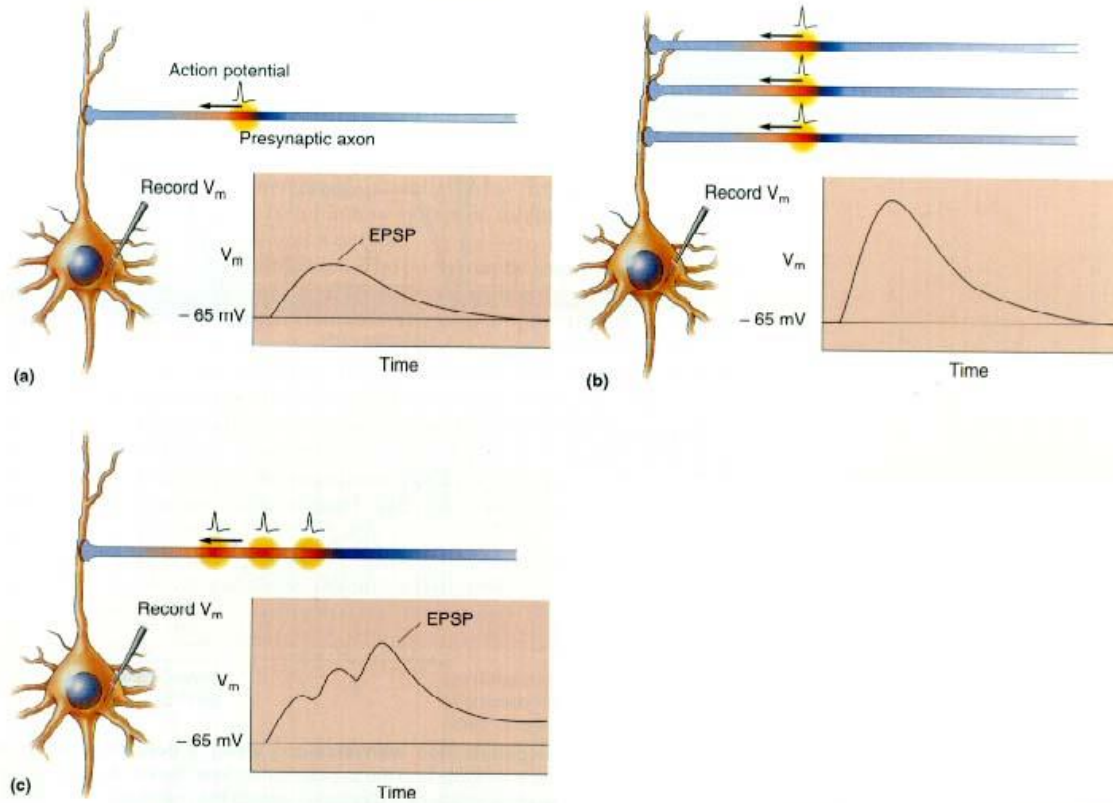
The postsynaptic potentials (PSPs) spread along the dendrites towards the soma and axon hillock where they can result in firing of an action potential.

PSPs can be excitatory (EPSPs, increase the probability of AP firing) or inhibitory (IPSPs, decrease it). The summation of a number of EPSPs is necessary to generate an action potential.



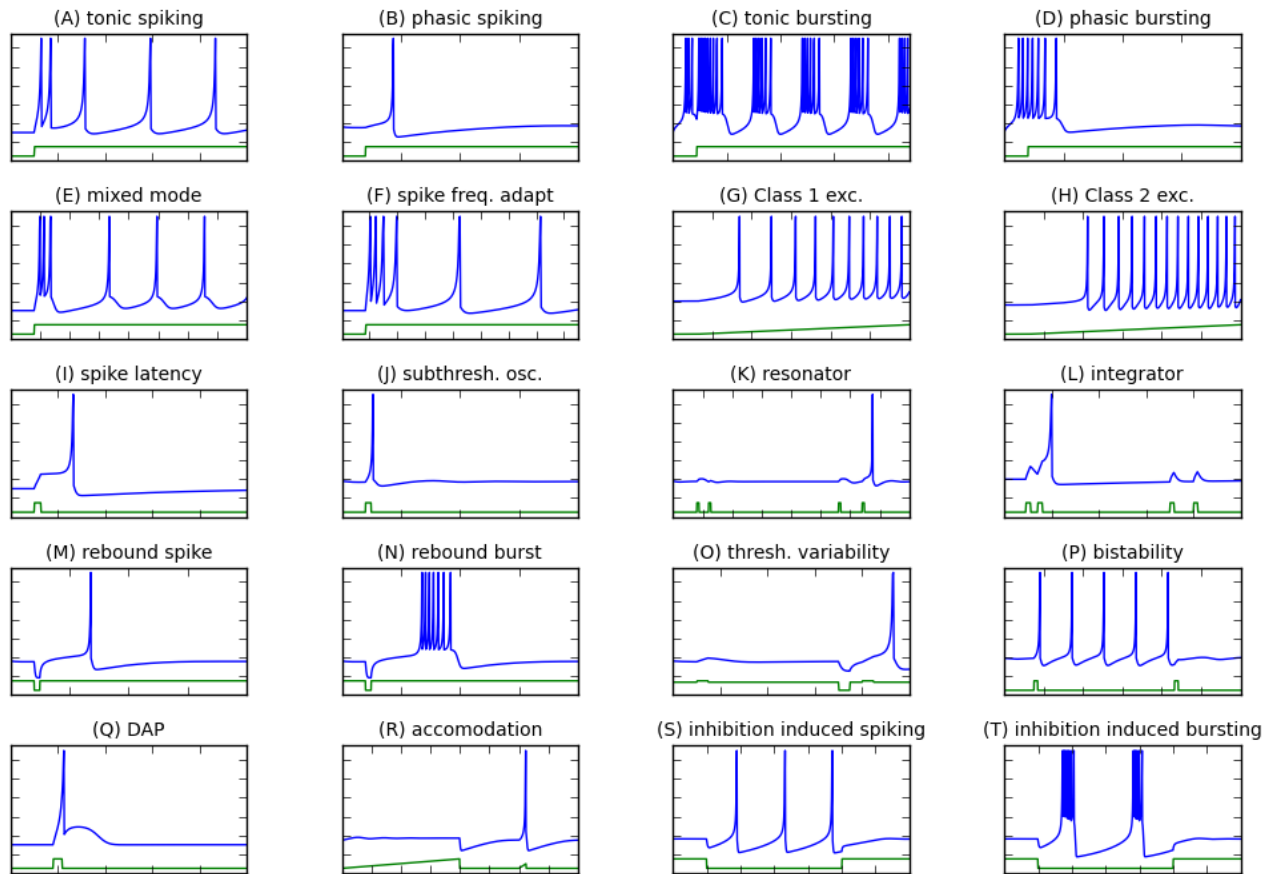
Spatial and temporal integration of PSPs

The summation of PSPs takes place both in space and time.



Neuronal firing patterns

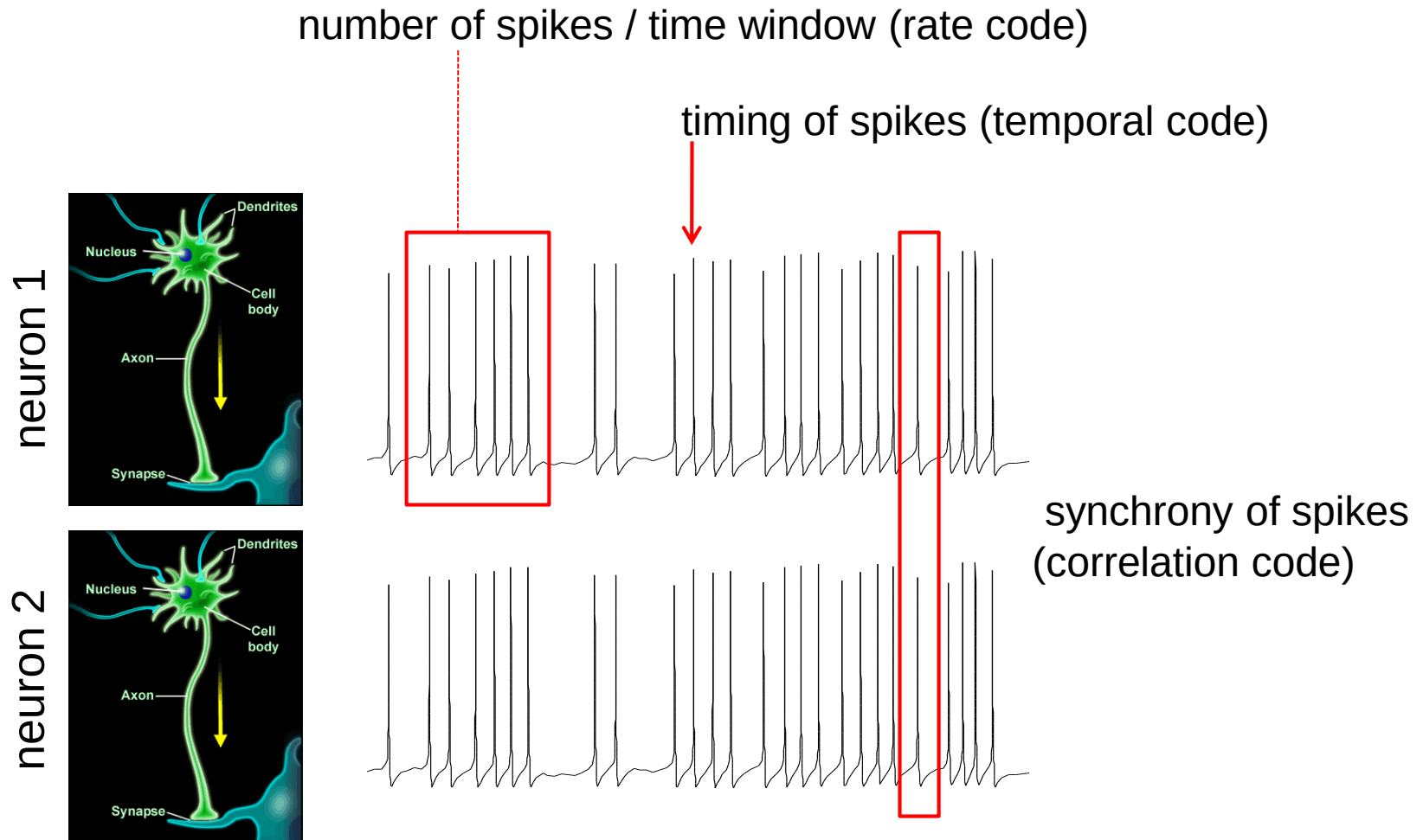
Different types of neurons fire different patterns of action potentials in response to synaptic input and application of electrical current.



How are these firing patterns used to transmit information?

Neural coding

The two main features of action potential trains that can be used to represent information are the mean rate of APs (rate coding) and the timing of APs (temporal coding).



Information can also be represented by synchrony of APs (correlation code).

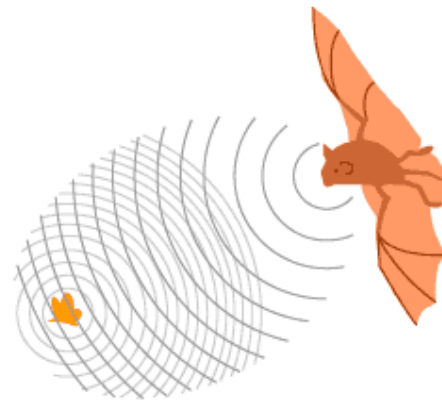
Examples for neural coding in biological systems

Rate coding

- Measurement of joint angles
- Detection of light intensity in the visual system

Temporal coding

- Binaural sound localization
- Echolocation in bats



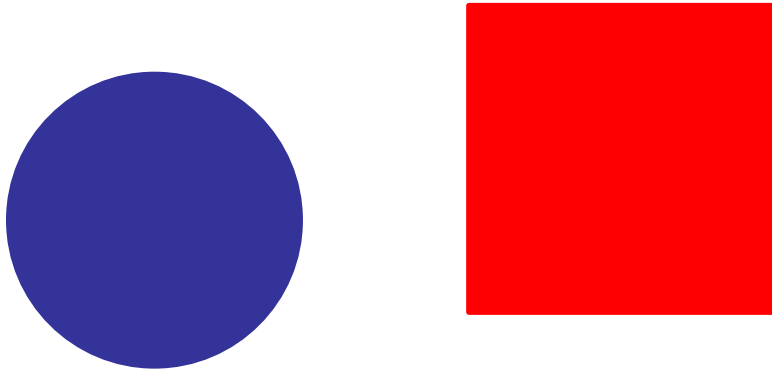
Correlation coding

- Odour recognition in insects
- Recognition of visual patterns, binding problem

Correlation coding and temporal coding are closely related: we can only measure the timing of a spike in correlation to another spike!

The binding problem

How does the brain represent that we look at a blue circle and a red square rather than a red circle and a blue square?

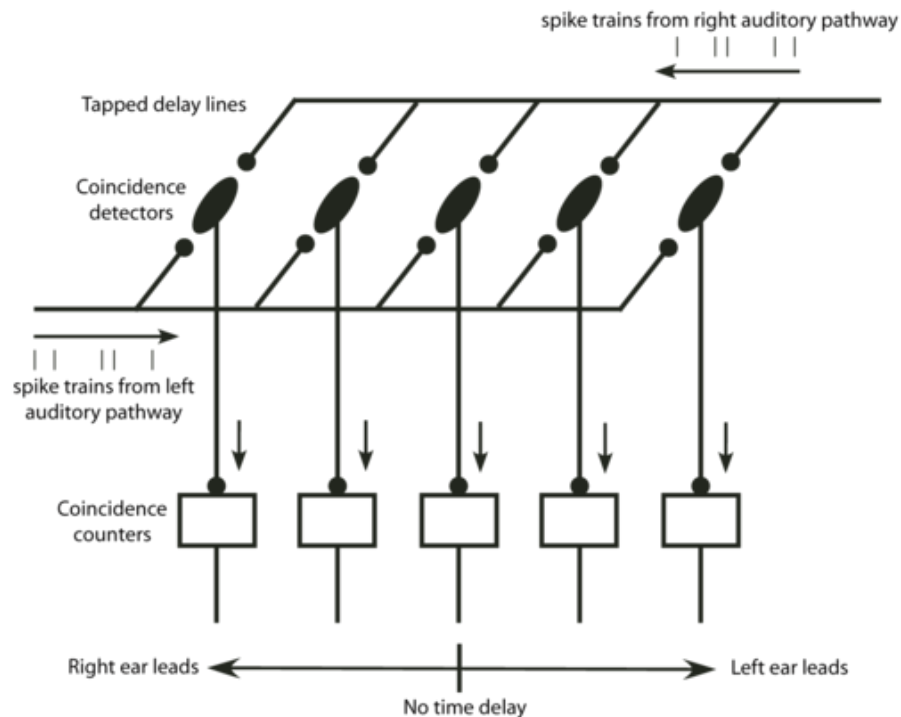


Hypothesis: the neurons representing “red” and “square” are synchronised, as are the neurons representing “blue” and “circle”.

Binaural sound localization: Jeffress model

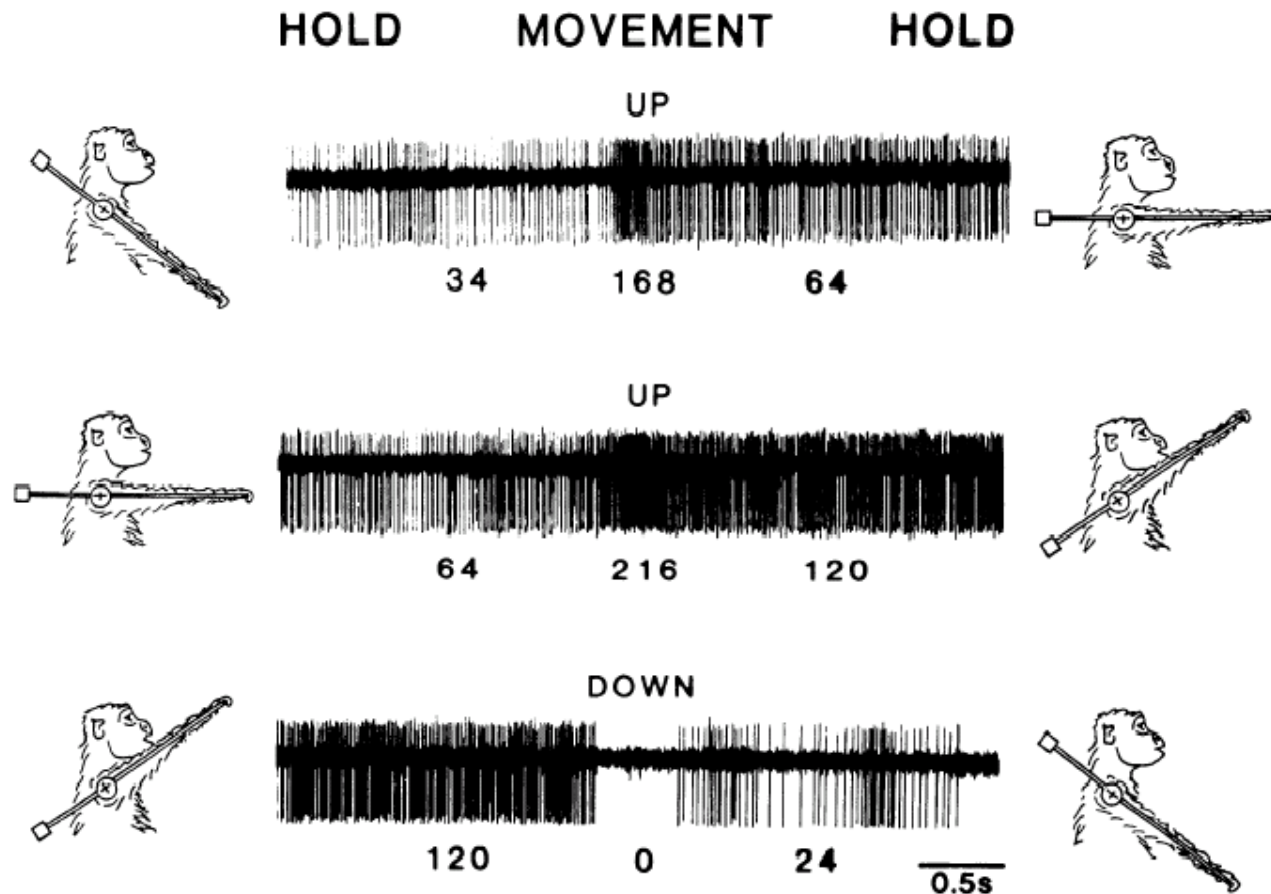
Jeffress (1948): the brain uses a combination of delay lines and coincidence detection to localise sounds in space.

Coincidence detectors: only respond when spikes from left and right ear arrive at the same time.



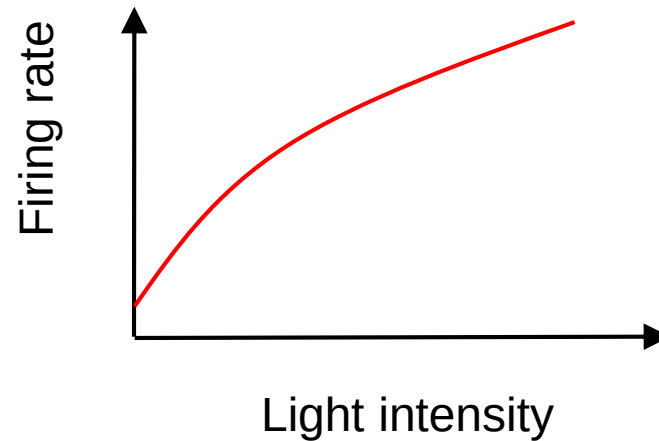
Measurement of joint angles

Van Kan et al. (1997): mossy fibres in the cerebellum use rate code to represent joint angles.

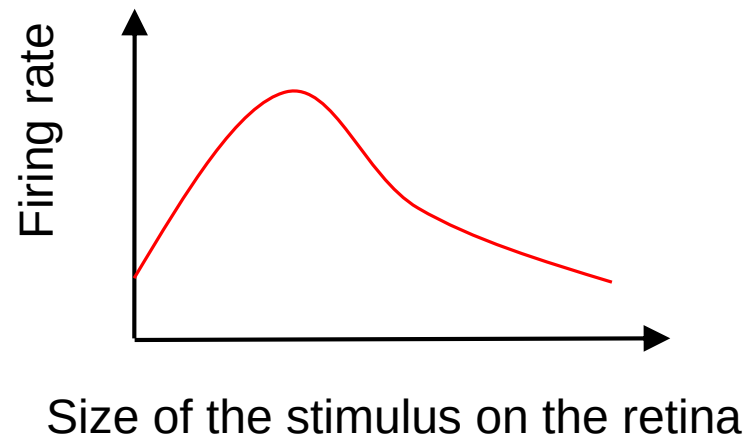


Response of ganglion cells in the retina

Light intensity is encoded by AP firing rate of retinal ganglion cells.

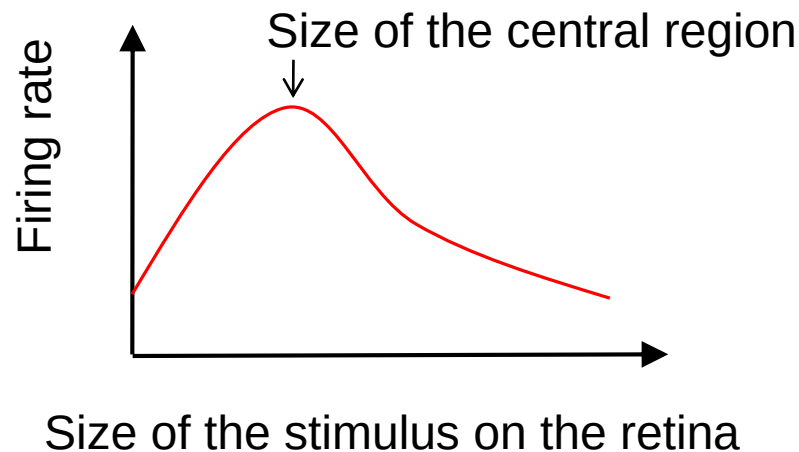
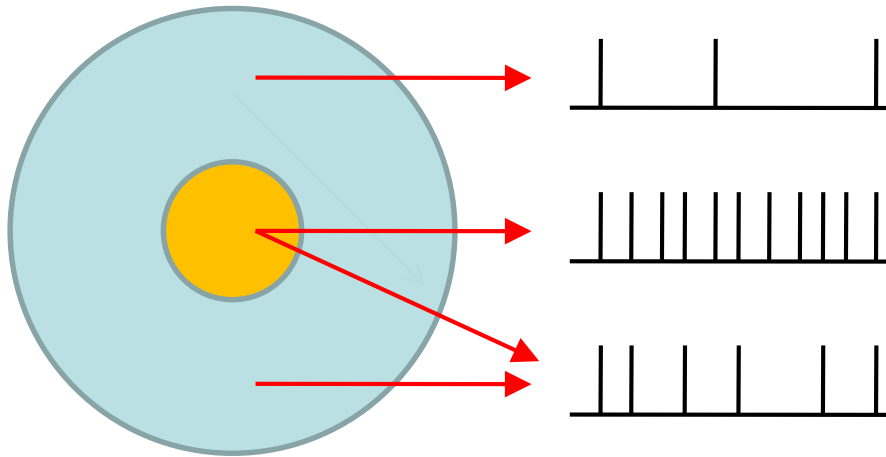


But the spatial structure of the stimulus also matters!



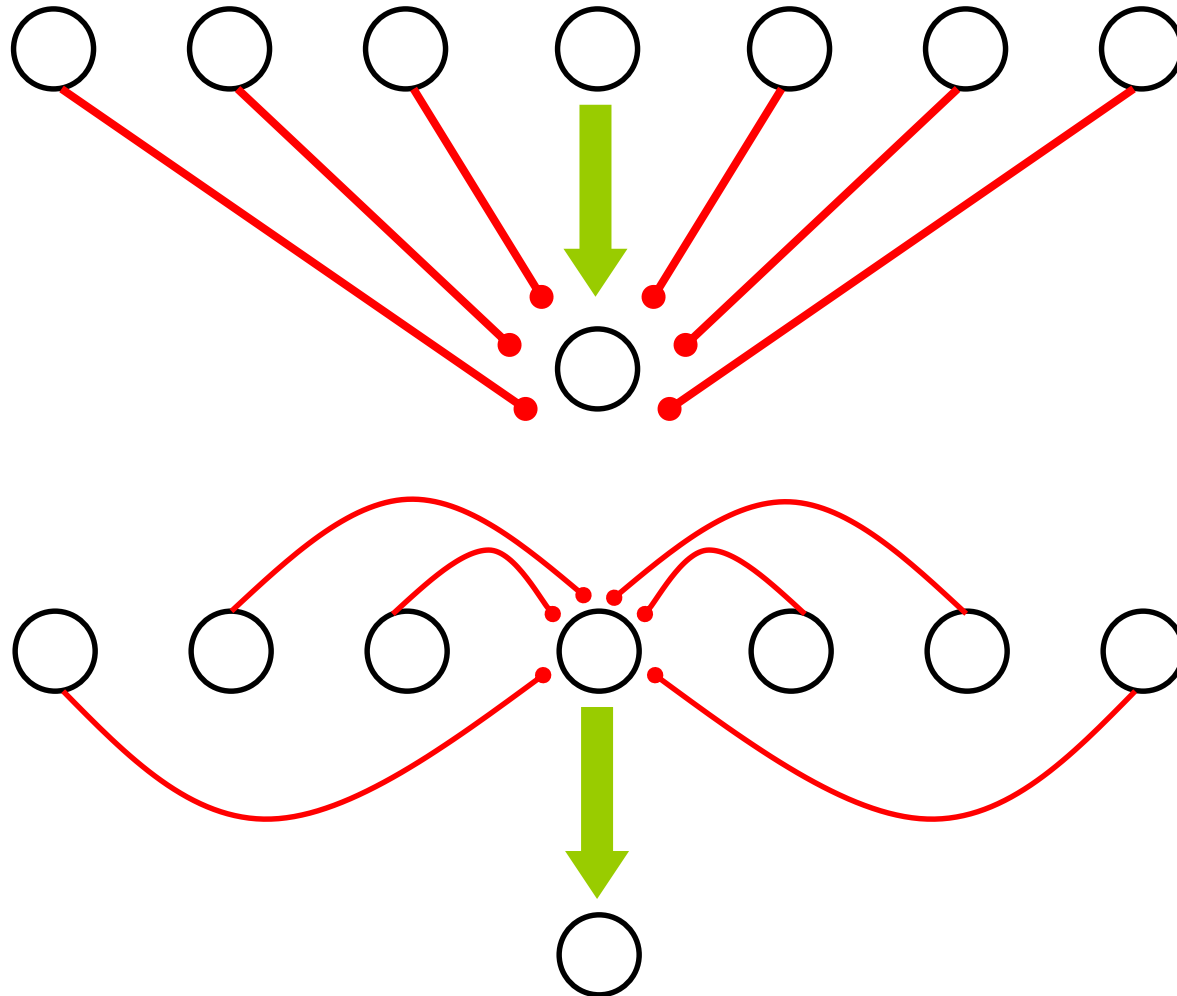
Receptive field of retinal ganglion cells

On-centre cells are activated by light in a central region and inhibited by light in a surrounding circle:

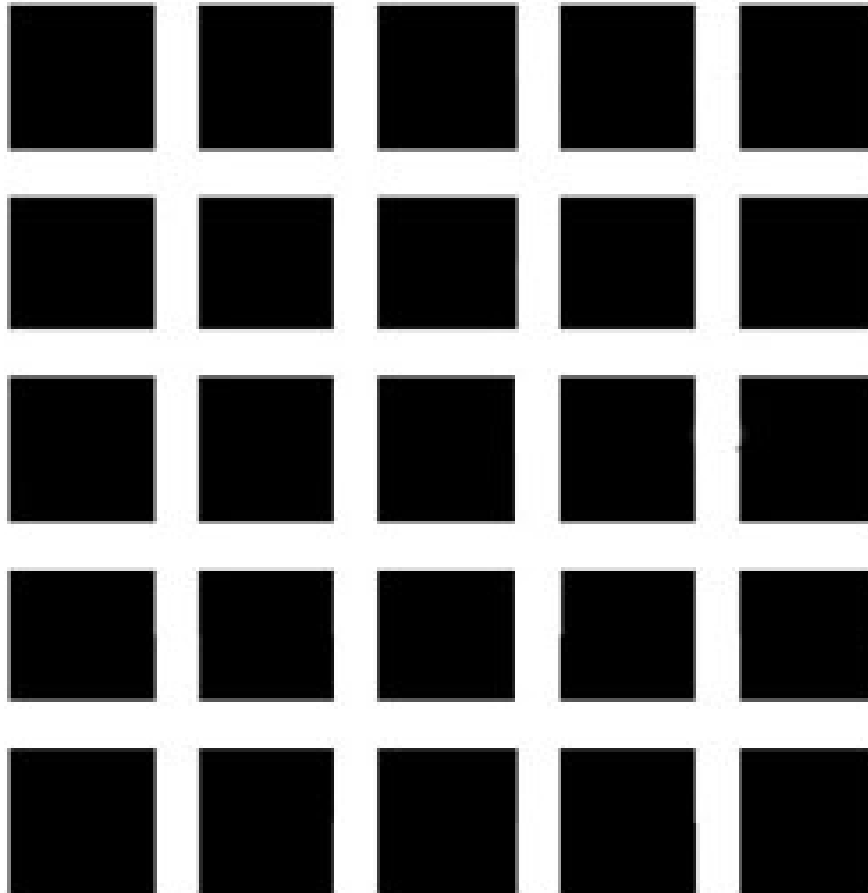


Neural circuitry underlying the on-center behaviour

Inhibitory surround can be implemented by lateral inhibition from neighbouring receptor cells. Two forms of lateral inhibition could be responsible:

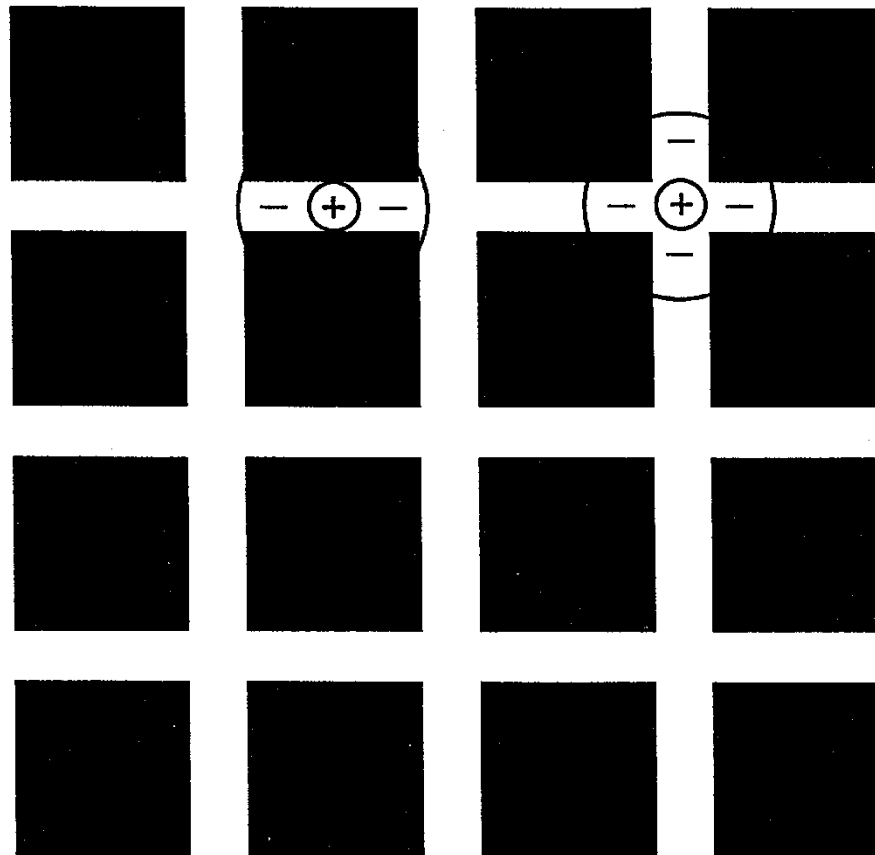


The Hermann Grid

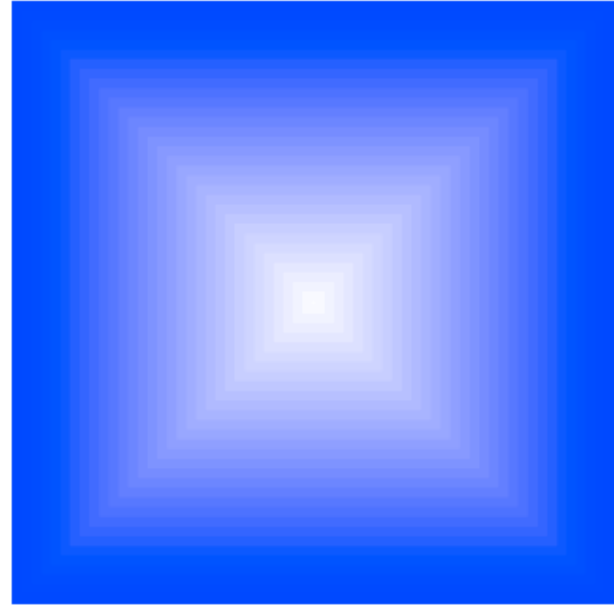
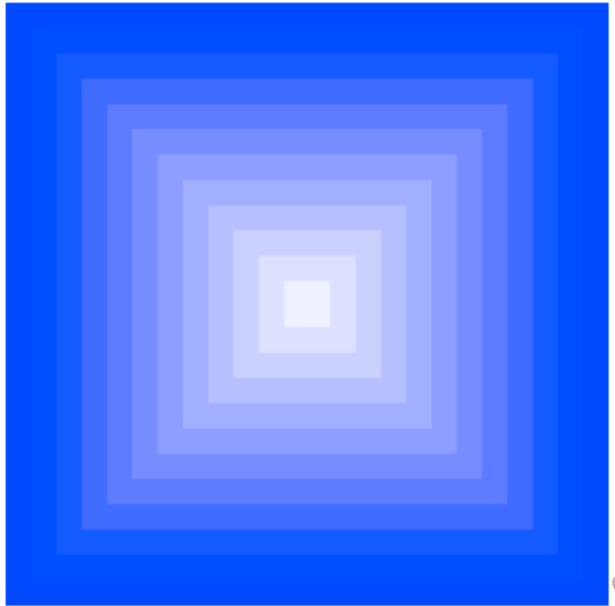


Explanation of the Hermann Grid

Optical illusion of grey spots at the crossing of white lines has been explained based on centre-surround receptive fields:

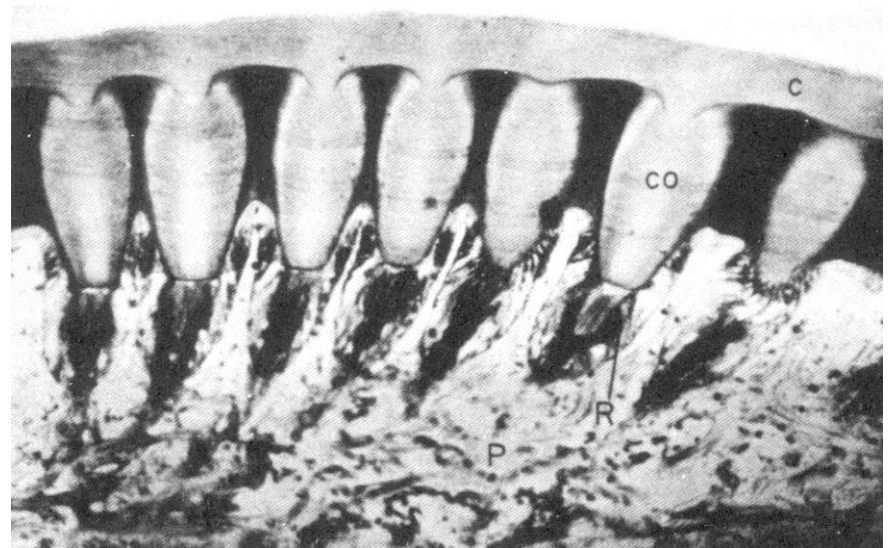
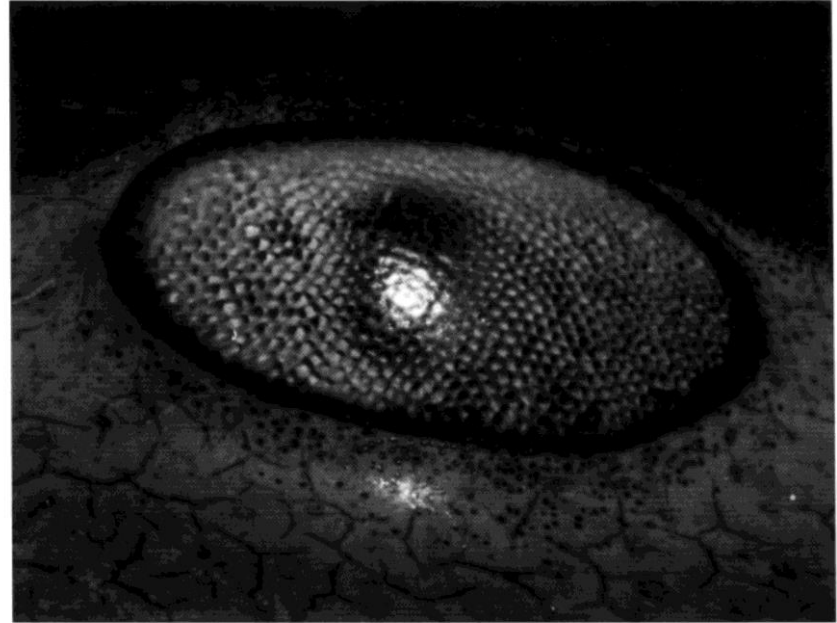


Vasarely paintings



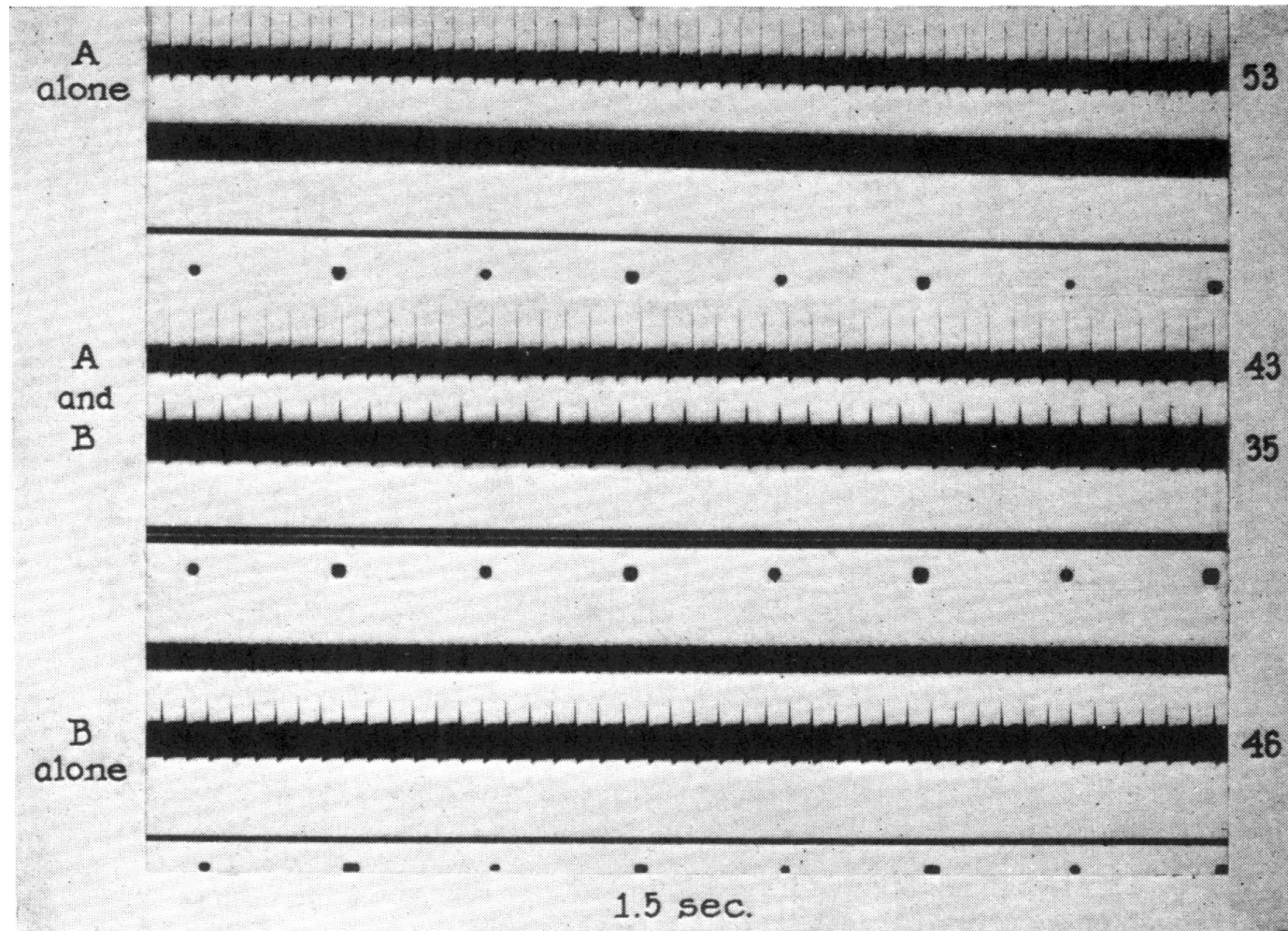
Lateral inhibition in the limulus eye

Limulus (horseshoe crab) eye comprises ommatidia that can be stimulated independently:
experiments by Ratliff & Hartline



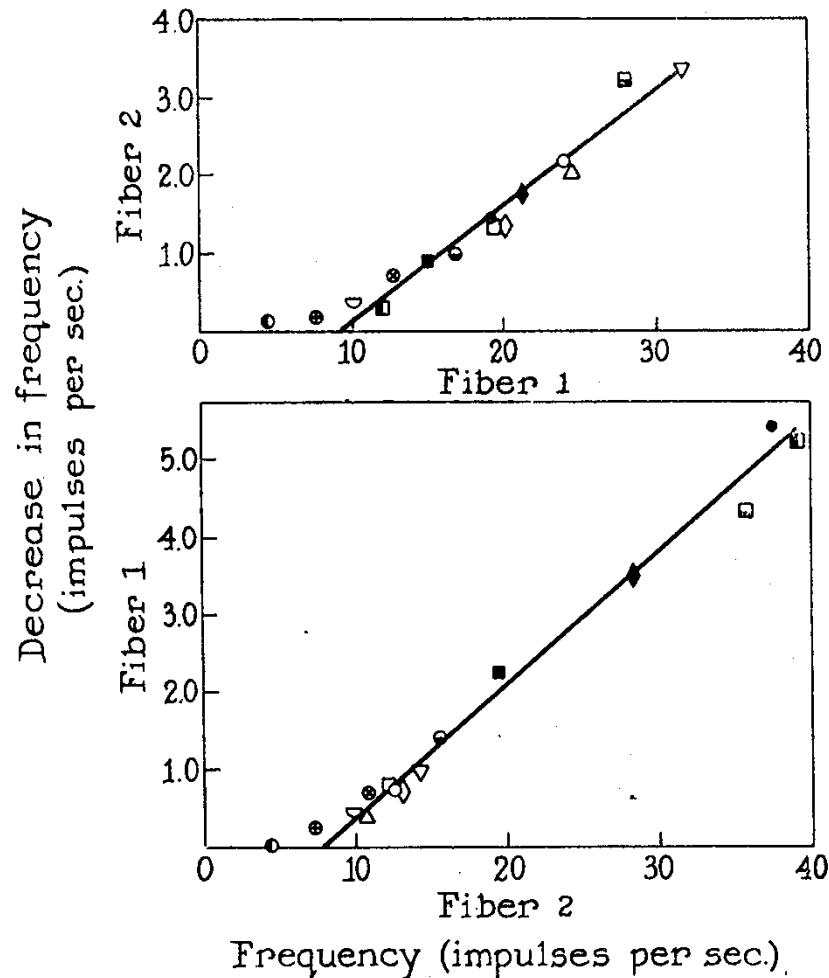
Stimulation of two ommatidia in the limulus eye

A composite stimulus (A and B) produces a sublinear response.



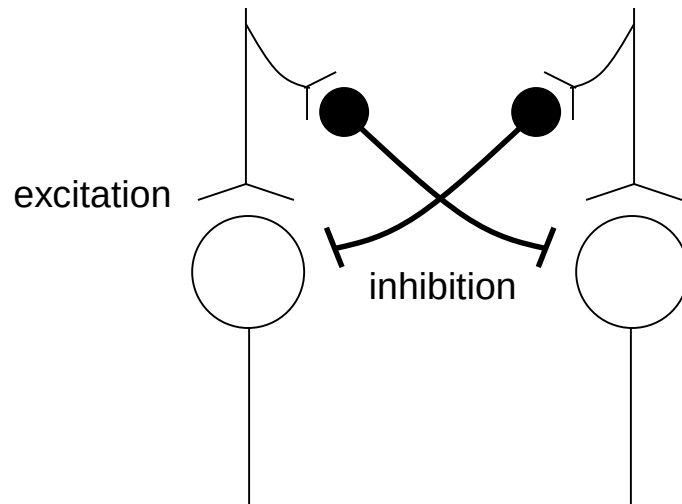
Neighbouring ommatidia inhibit each other

The decrease in firing rate of each ommatidium increases with the firing rate of the neighbouring ommatidium.

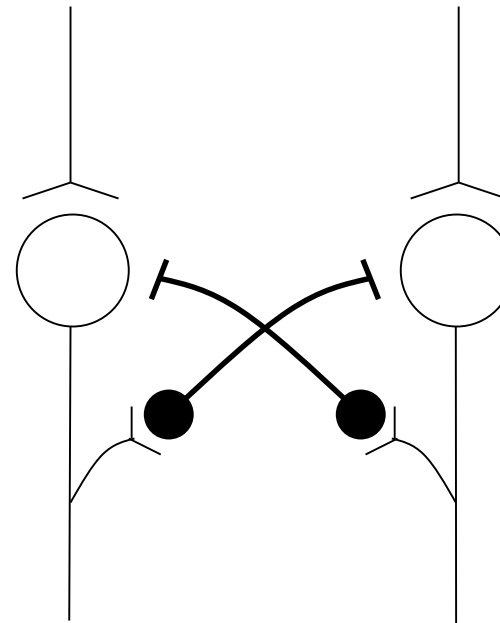


Two possible mechanisms for lateral inhibition

Lateral inhibition could be implemented by feed-forward or feedback circuits.



feed-forward



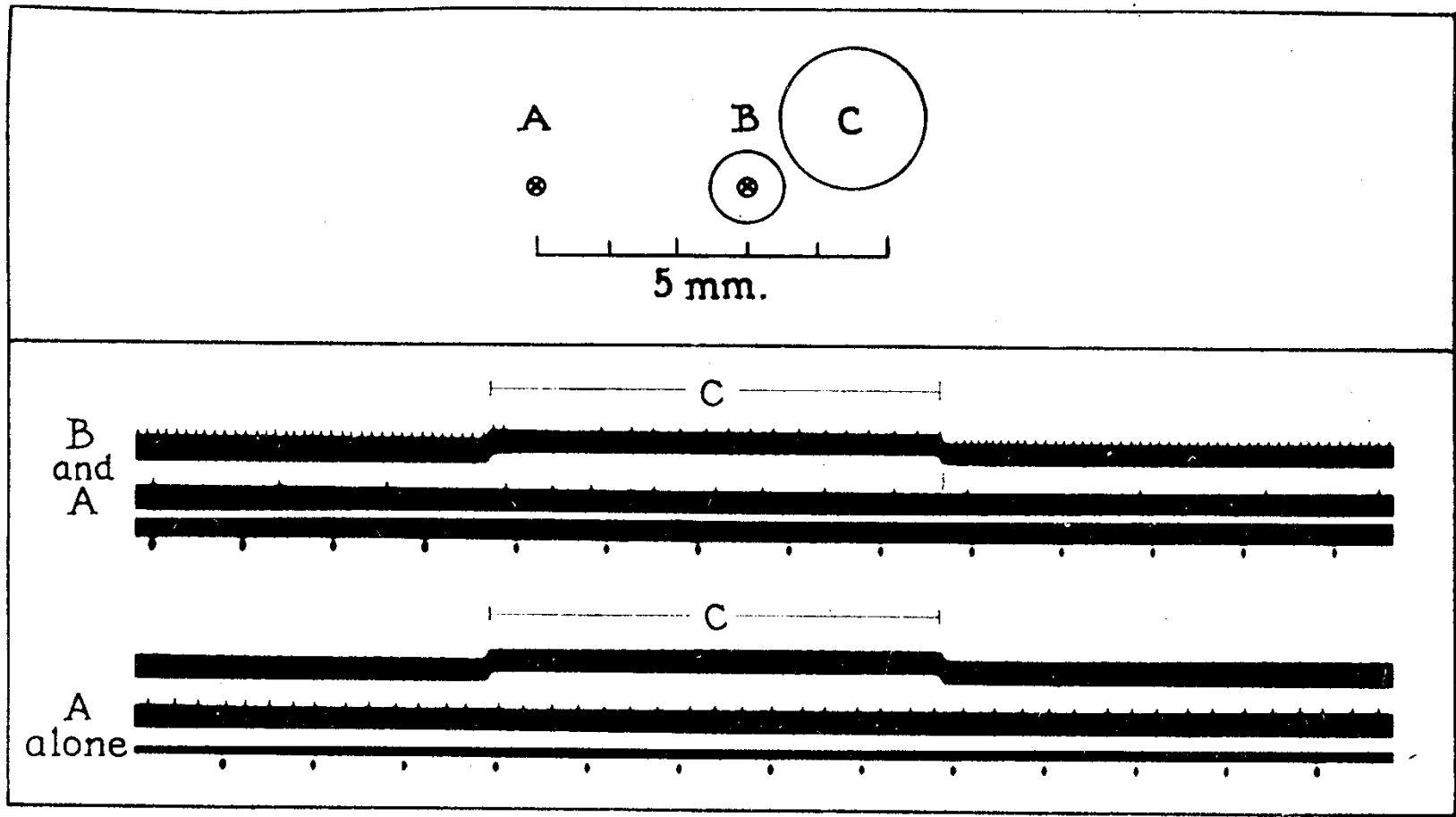
feedback

flow of information

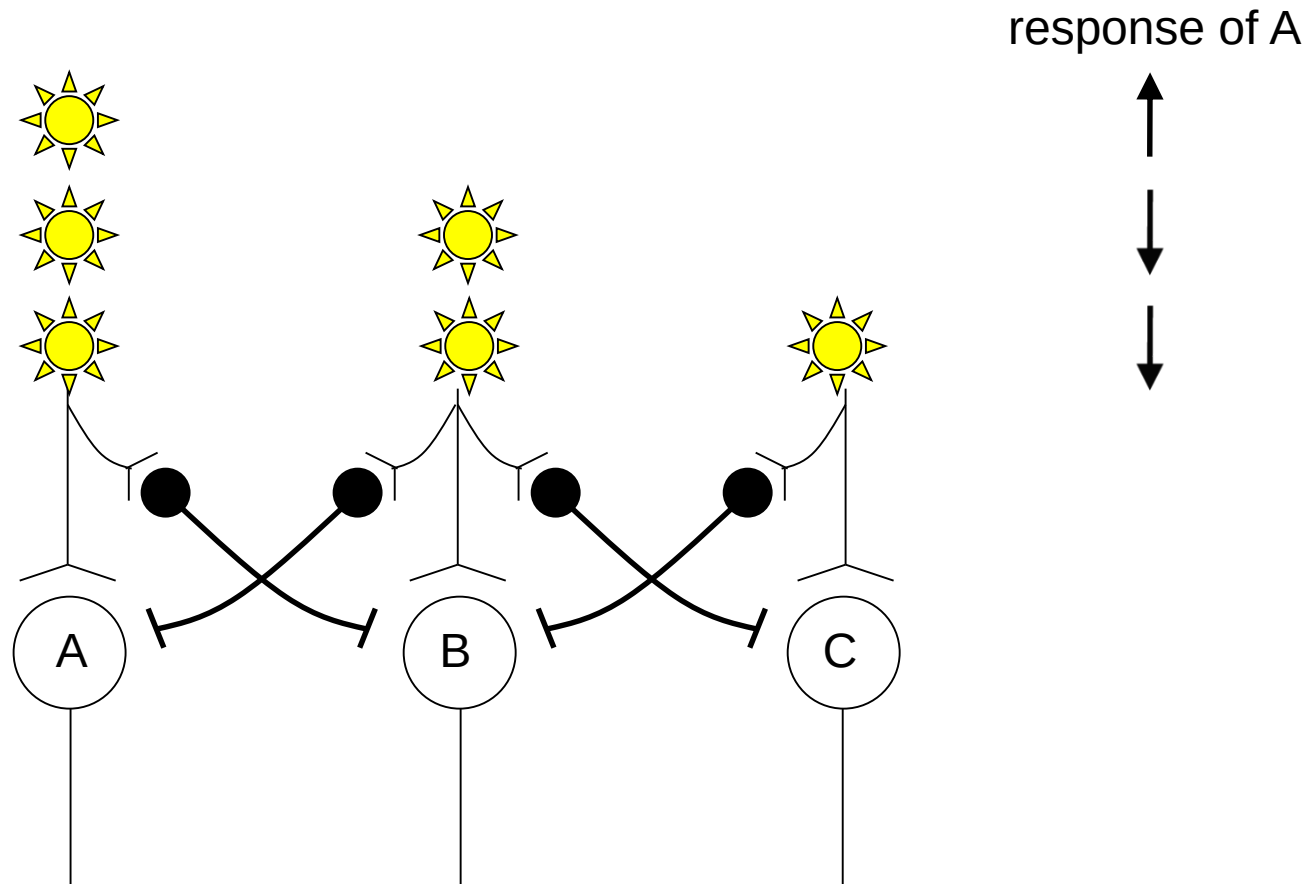


Stimulation of a third ommatidium

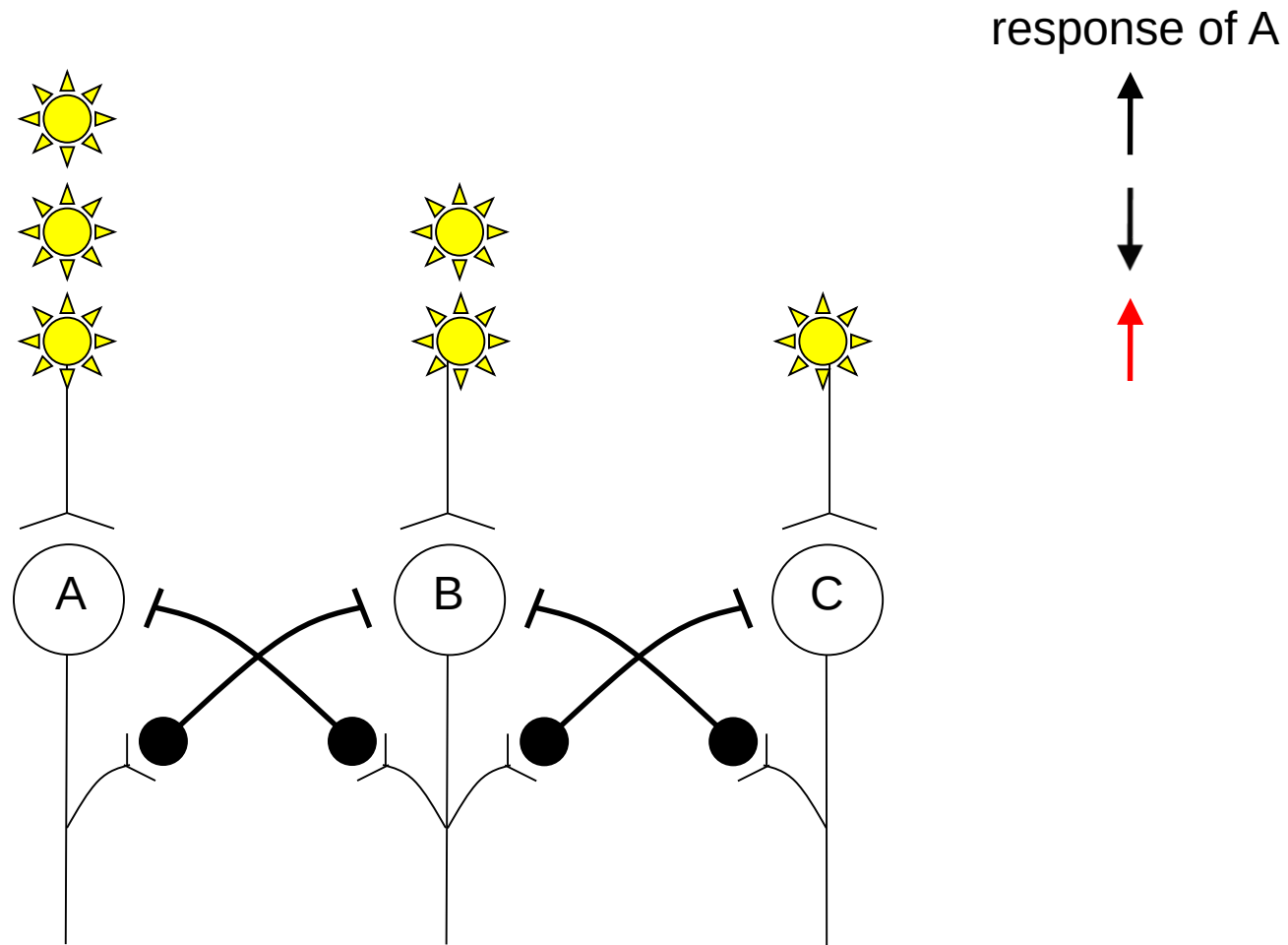
The stimulation of a third ommatidium reveals that the lateral inhibition is implemented by a feedback circuit.



Expected response in a feed-forward network



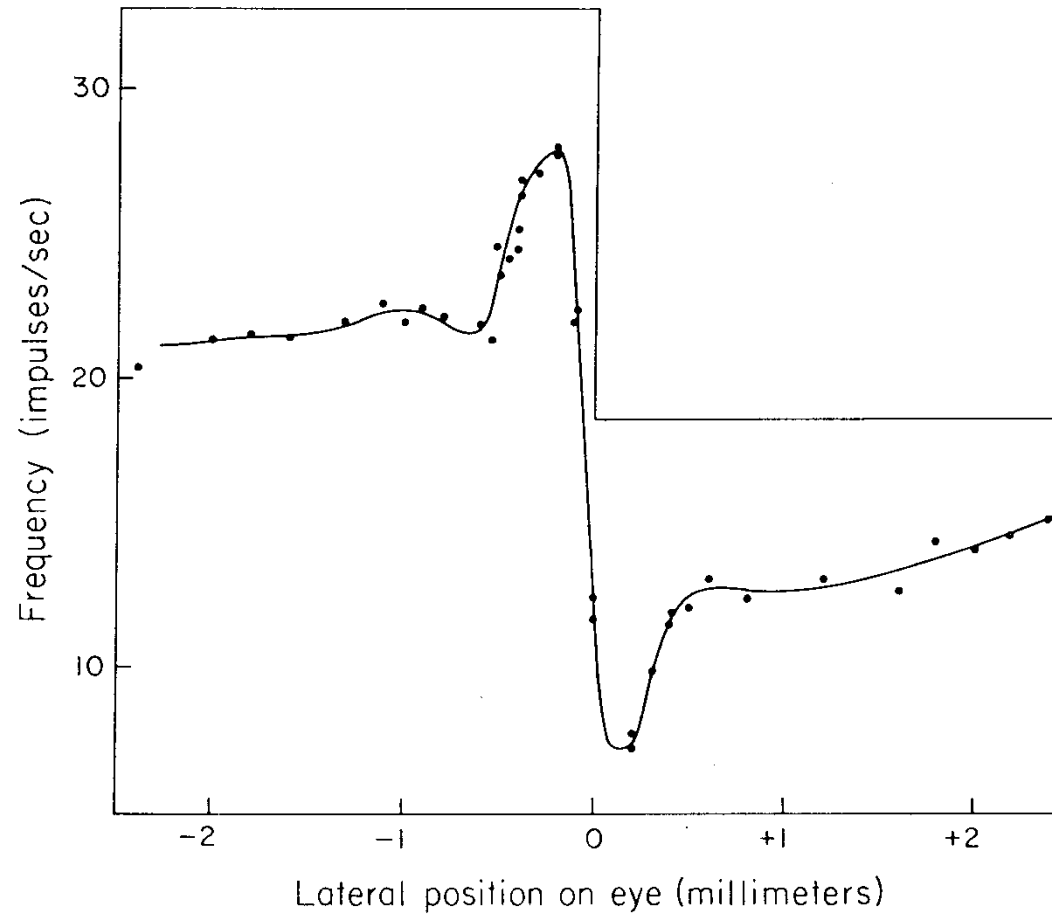
Expected response in a feedback network



As this is what Ratliff and Hartline observed, they concluded that lateral inhibition in the limulus eye is mediated by a feedback circuit.

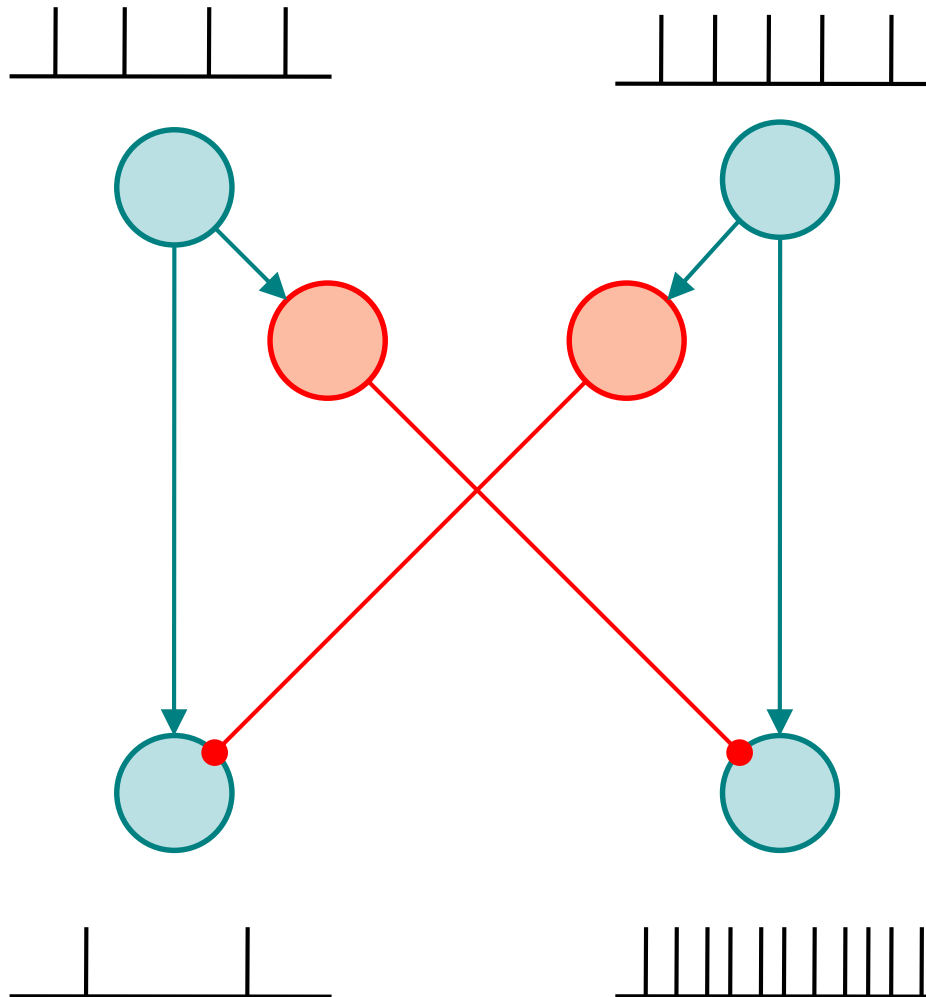
Contrast enhancement by lateral inhibition

Response of the limulus eye to a black/white boundary:



General functions of lateral inhibition (1)

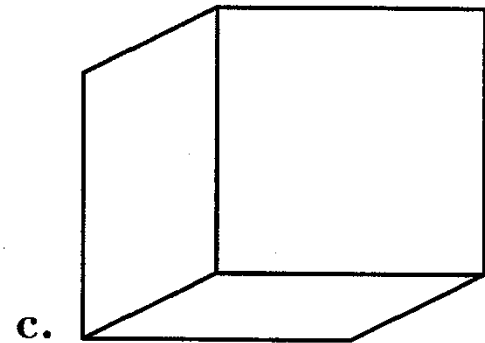
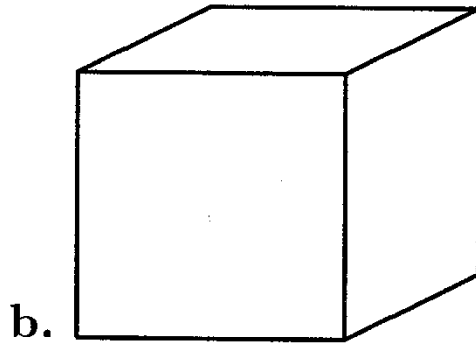
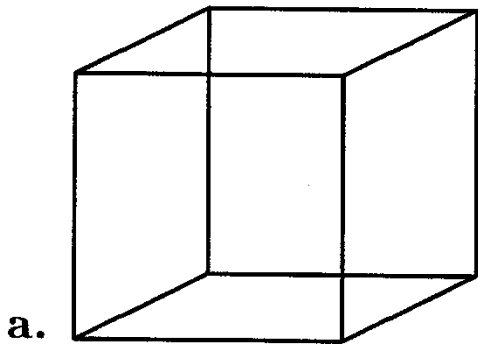
Lateral inhibition is an important computational principle as it amplifies the differences between inputs and increases the sensitivity of the system.



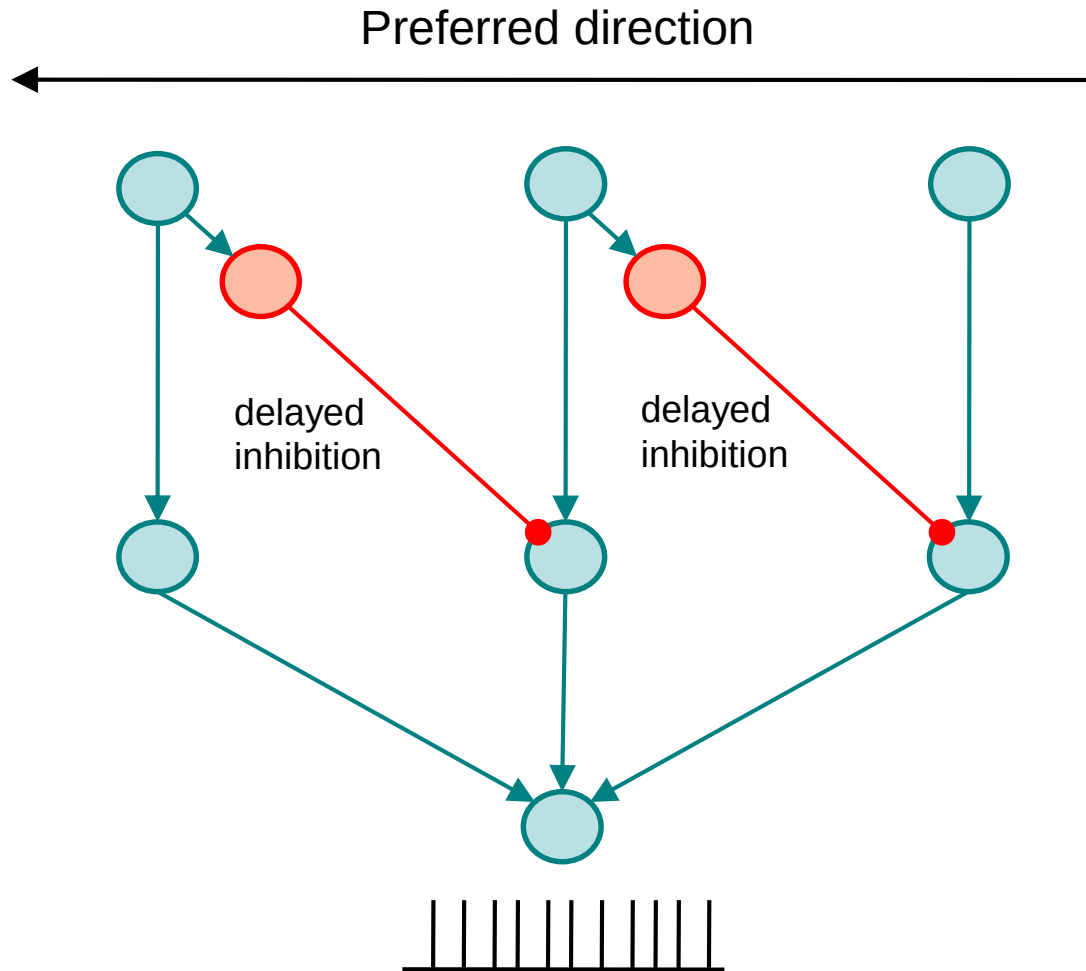
General functions of lateral inhibition (2)

Lateral inhibition can result in winner-take-all (WTA) responses. This is widely applied in artificial neural networks.

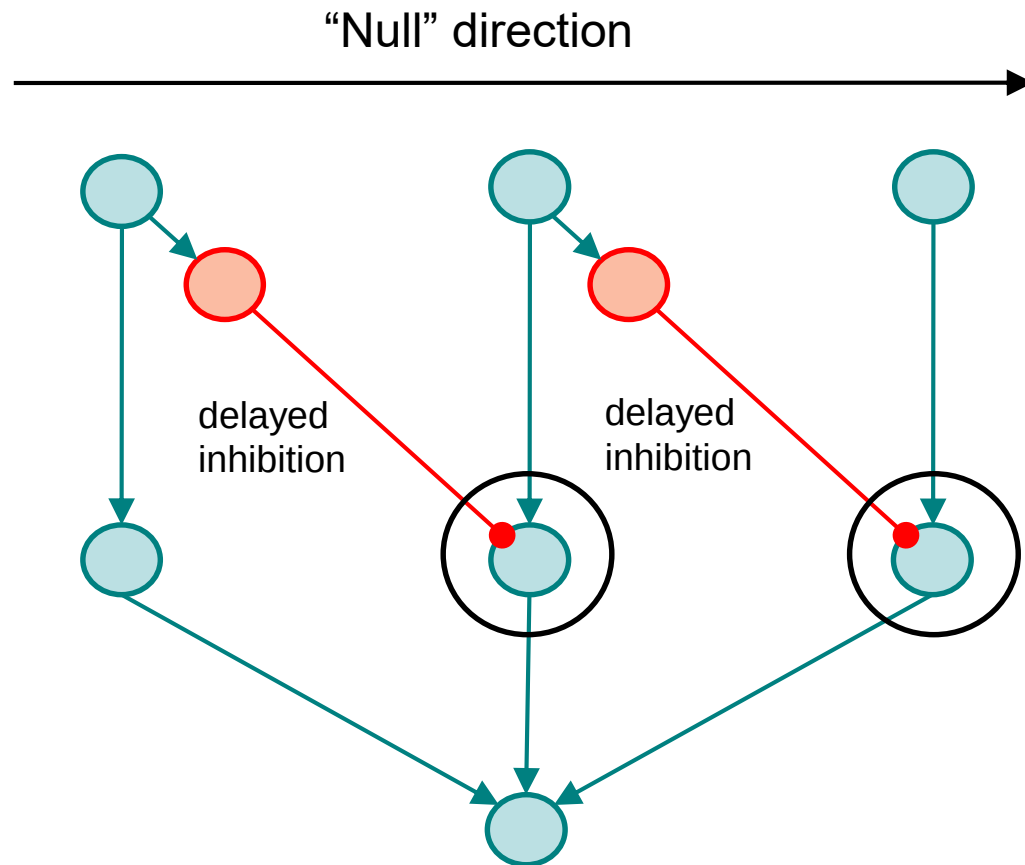
Example for WTA in perception: Necker cube.



Directed inhibition can implement direction selectivity



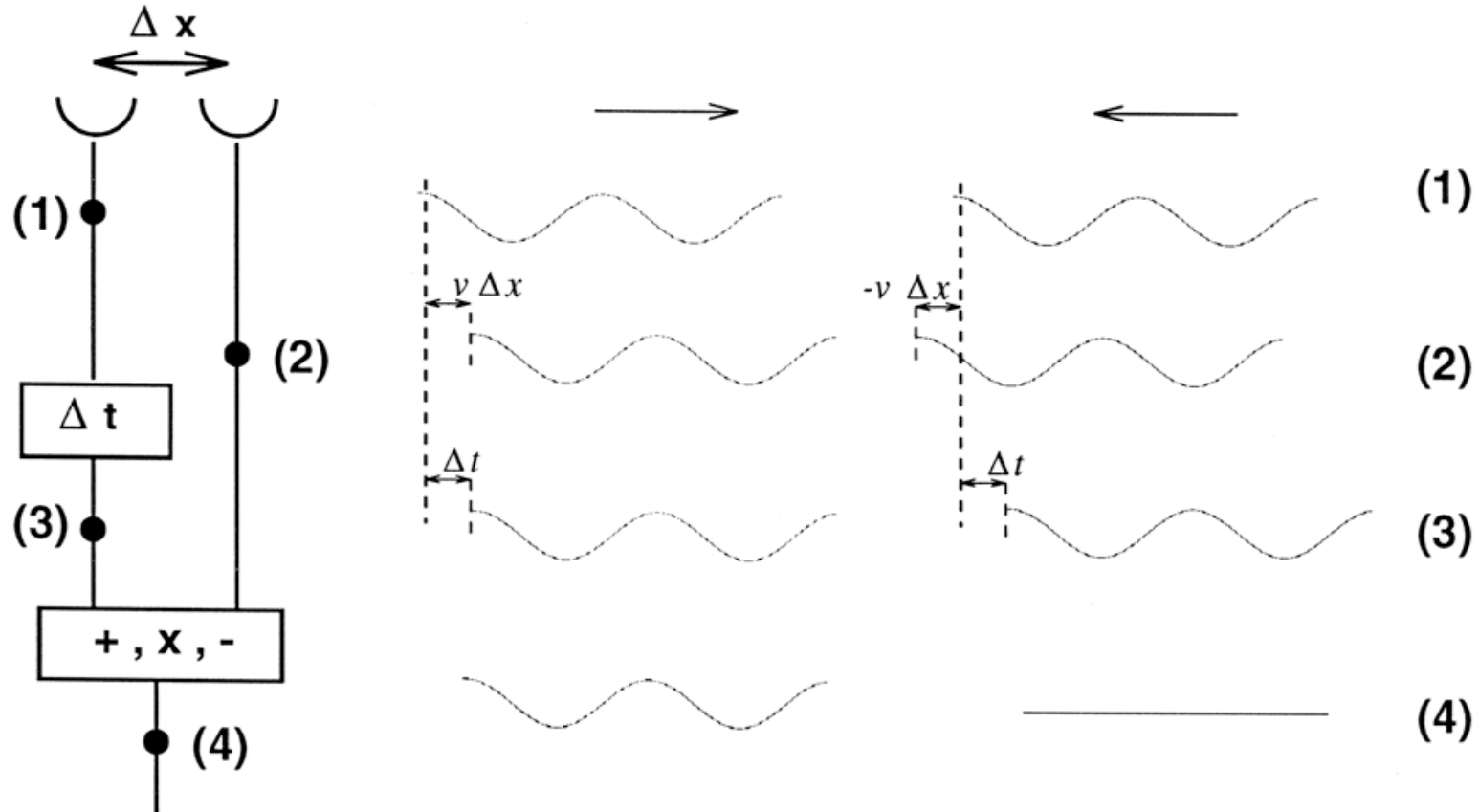
Direction selectivity based on inhibition (2)



No response as the delayed inhibition cancels out the excitation from the next sensor in the sequence

Direction selectivity without inhibition

Reichardt detector (1969): suggested as basis of motion detection in flies.

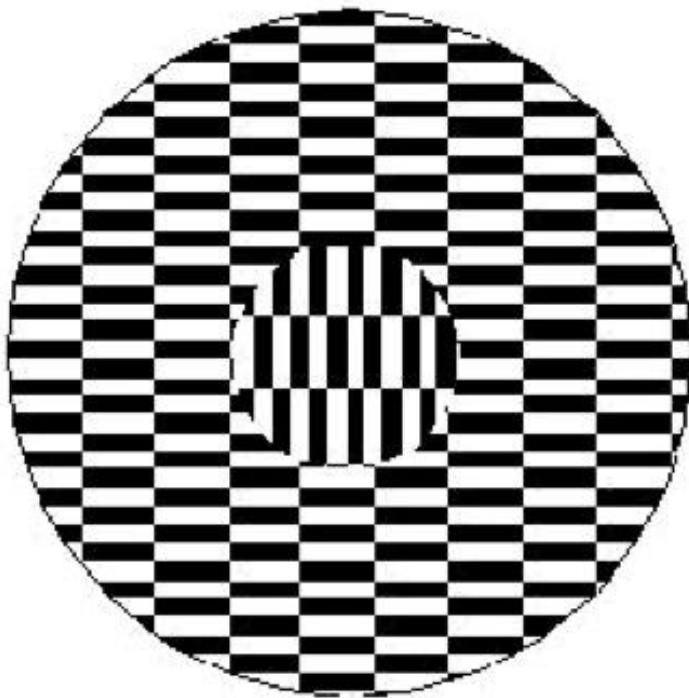


Two light detectors (1) and (2), separated by a distance Δx , feed into a correlator (4). One of the signals is delayed by Δt (3). Difference to motion in opposite directions is greatest when the stimulus moves at speed $\Delta x / \Delta t$.

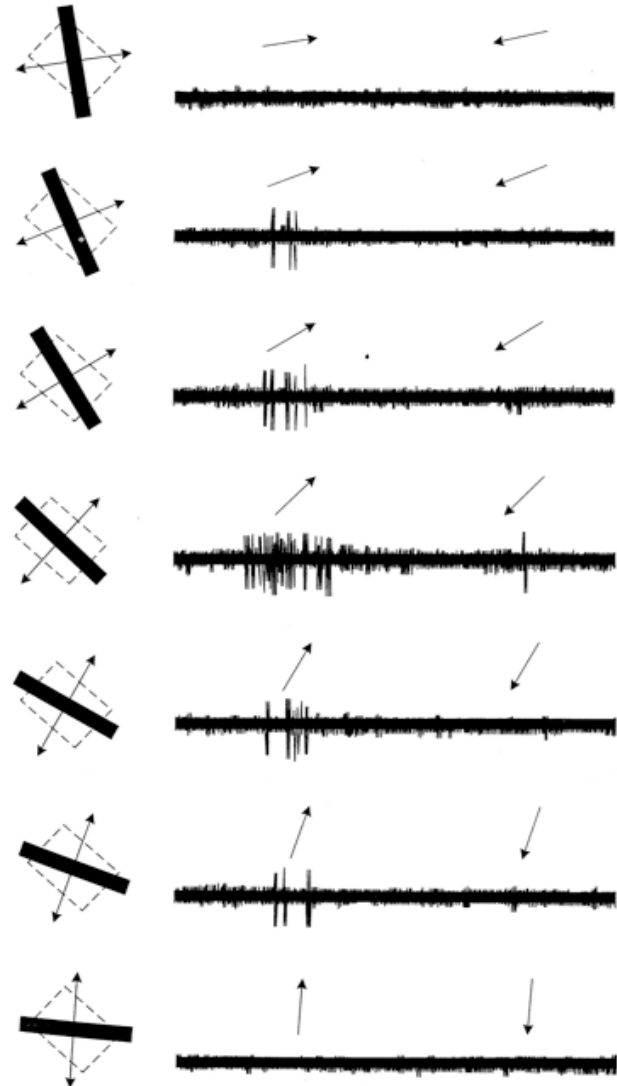
Direction selective responses in visual cortex

The mammalian visual cortex also contains orientation and direction selective neurons.

Ouchi illusion: has been explained by the preferred detection of motion orthogonal to straight edges:



moving bar neuronal response



Modifiable synapses and learning

Until now we have assumed that biological neural circuits are hardwired. In fact, most synapses are plastic and update their weights depending on the activity of the pre- and postsynaptic neuron.

Moreover, many neurons continuously form new connections and eliminate existing ones.

Synaptic plasticity is important as it forms the basis of learning and memory.

Learning in real and artificial neural networks can be divided into two main categories: supervised learning and unsupervised learning

Supervised and unsupervised learning

Supervised learning

- “Learning with a teacher”
- The desired learning outcome is known
- For each input, there is a predetermined target output
- Learn associations between patterns (input_i , output_i)
- Labelled data

Unsupervised learning

- Learning by self-organisation
- Unlabelled data, no set targets
- Detect regularities in the inputs
- Requires redundancy in the data

Can we find examples for the two types of learning in the brain?

Supervised learning: eye-blink conditioning

Eye-blink reflex: we react to an airpuff or a shock to the eye with an eye-blink.

Classical conditioning of the eye-blink reflex: we can learn to associate the eye-blink with another stimulus such as a tone or light.

Terminology:

- shock = unconditioned stimulus (US)
- tone = conditioned stimulus (CS)
- eye-blink in response to shock = unconditioned response (UR)
- eye-blink in response to tone = conditioned response (CR)

After several paired presentations of CS and US, the CS will trigger a CR.

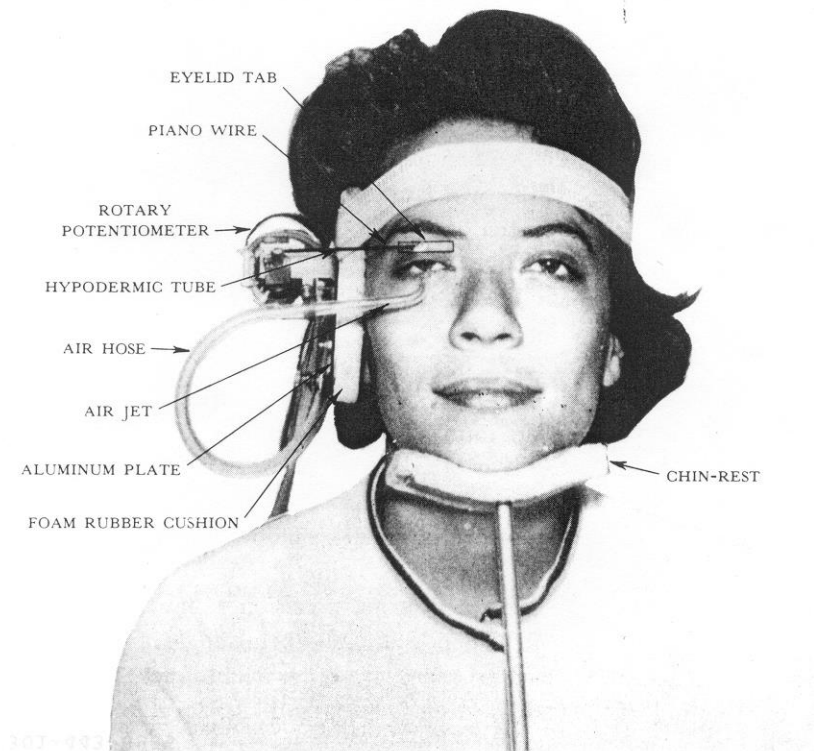
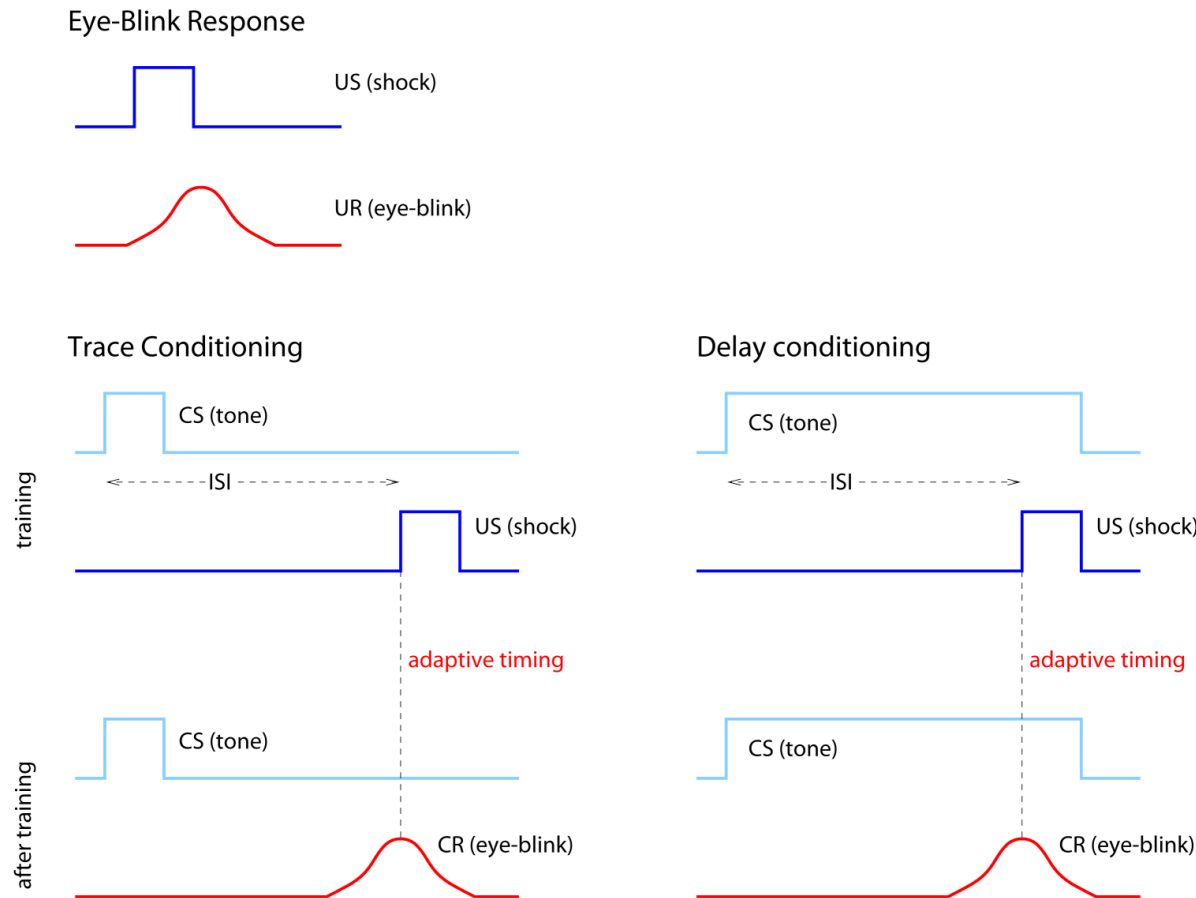


Fig. 9-3 A subject wearing the headgear for recording eyelid movement.

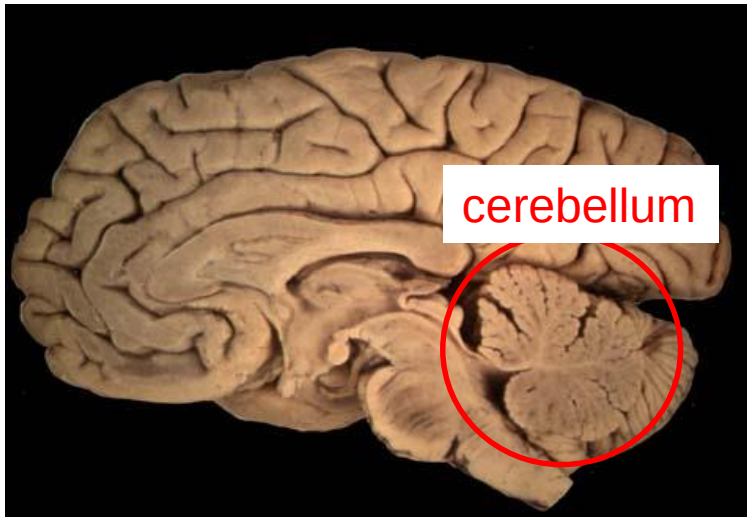
Classical conditioning of the eye-blink response

The CS and US can be separated in time (trace conditioning) or overlap (delay conditioning).



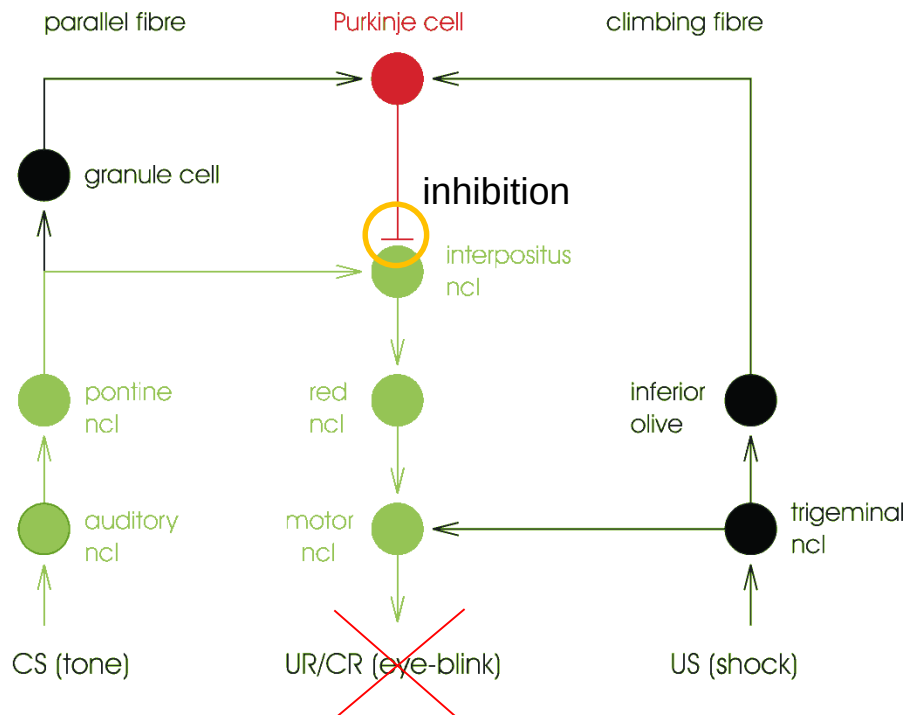
The CR occurs at the time of US presentations during training (adaptive timing).

The neural basis of eye-blink conditioning



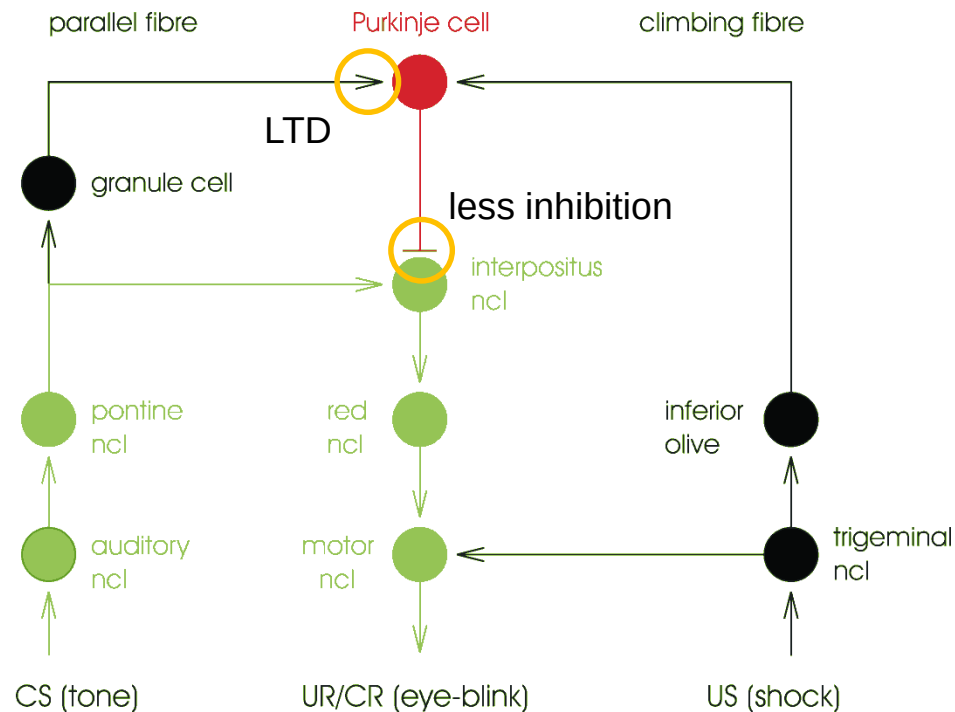
Eye-blink conditioning is mediated by the cerebellum, a part of the brain that is involved in motor control.

The sensors that respond to the tone CS can in principle activate the muscles that trigger the eye-blink CR: there is a connection through different brain nuclei (indicated in green).



Before training, this connection is inhibited by input from Purkinje cells in the cerebellum.

Synaptic plasticity implements eye-blink conditioning



During conditioning, paired presentations of CS and US result in the paired activation of two inputs to Purkinje cells: parallel fibre and climbing fibre input.

This induces synaptic plasticity: the synapses between parallel fibres and Purkinje cells are weakened by a process called Long-term Depression (LTD).

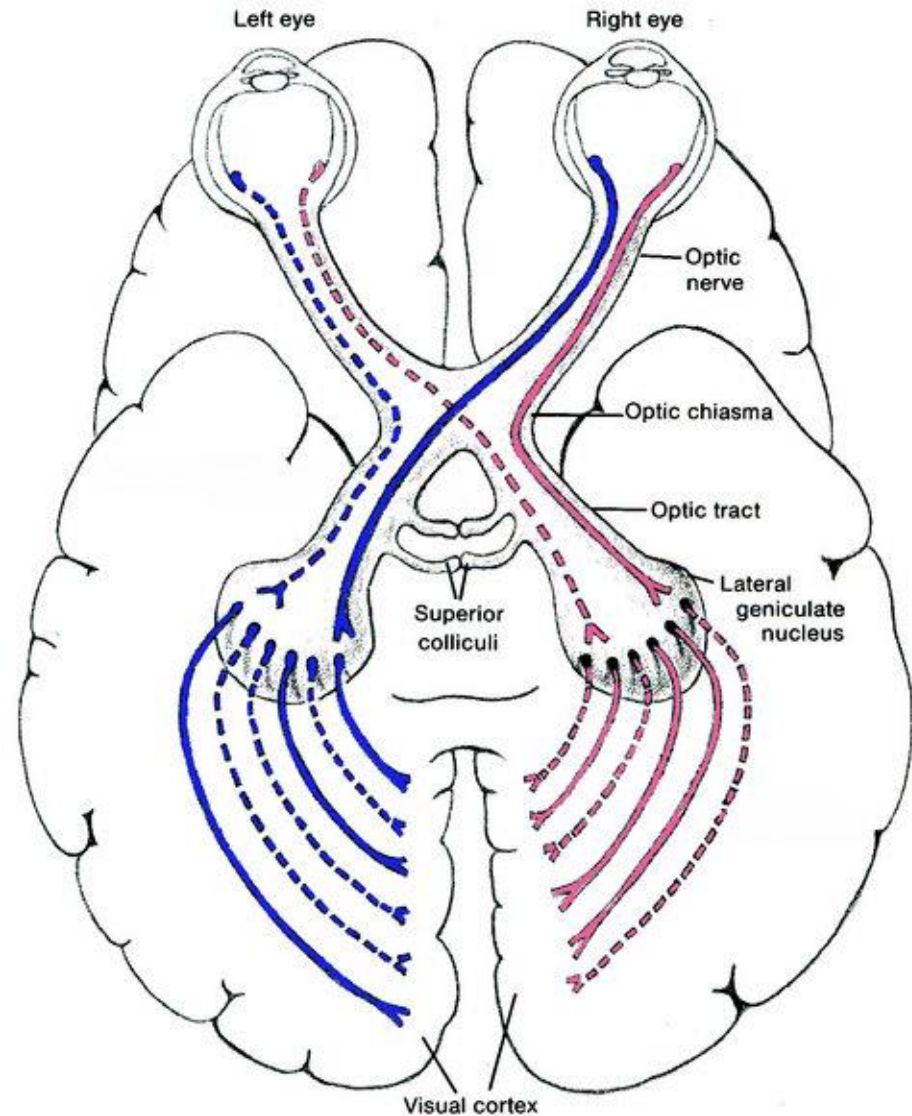
As a result, the Purkinje cells will be activated less and the connection between CS and CR will be inhibited less: a CS can now trigger a CR.

Unsupervised learning: development of the visual cortex

Information from both eyes is transmitted to the primary visual cortex.

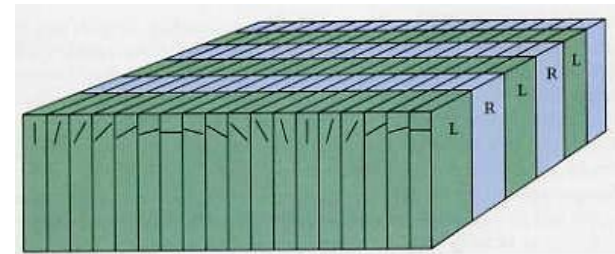
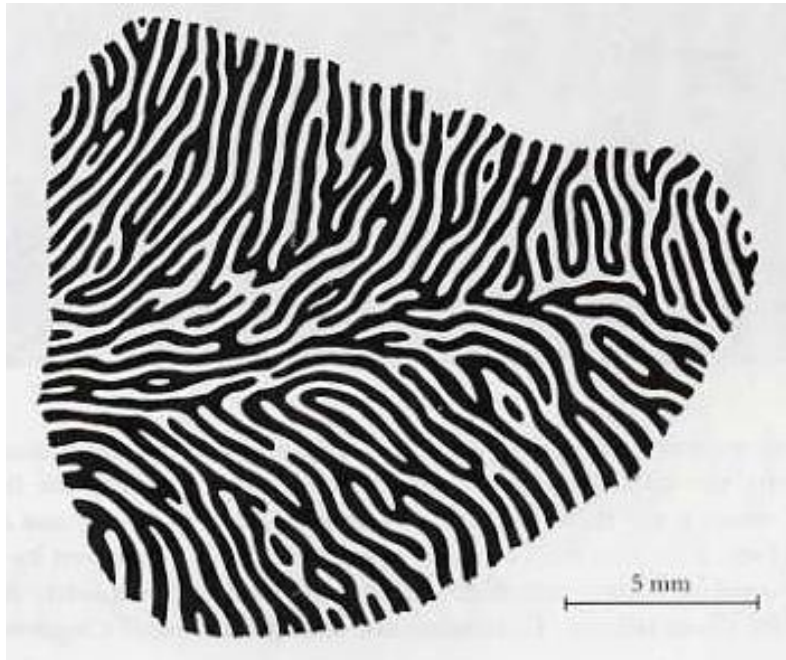
The left (right) cortical hemisphere receives input from the right (left) visual field.

Both hemispheres receive input from both eyes.



Ocular dominance columns in the visual cortex

Inputs from the right and left eye activate alternating columns of neurons in primary visual cortex.



The development of these ocular dominance columns depends on input from the eyes during a critical period (4 months in cats, 12 months in monkeys).

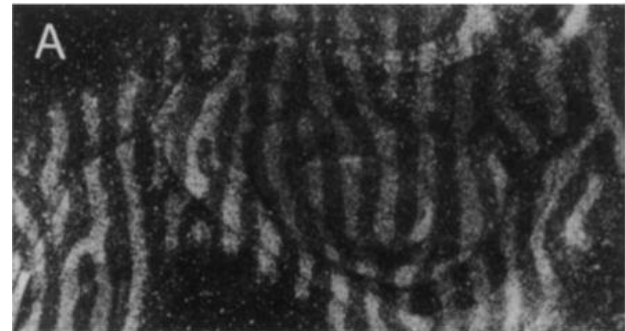
Monocular deprivation experiments

Hubel and Wiesel found that closing one eye during the critical period resulted in a drastic reduction of the ocular dominance columns that received input from the closed eye.

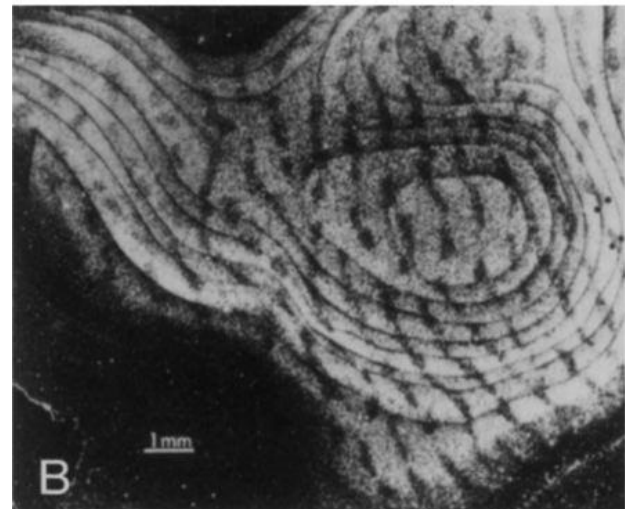
This has implications for the treatment of strabismus in humans: it needs to be corrected early, otherwise blindness in one eye can result as the dominant eye takes over.

The formation of ocular dominance columns is an example for unsupervised learning in the brain.

Other examples include the development of motion and direction selective cells. Kittens that are reared in an environment comprising only horizontal stripes will not be able to see vertical stripes as adult cats.



normal



deprived

The brain can be studied at many different levels

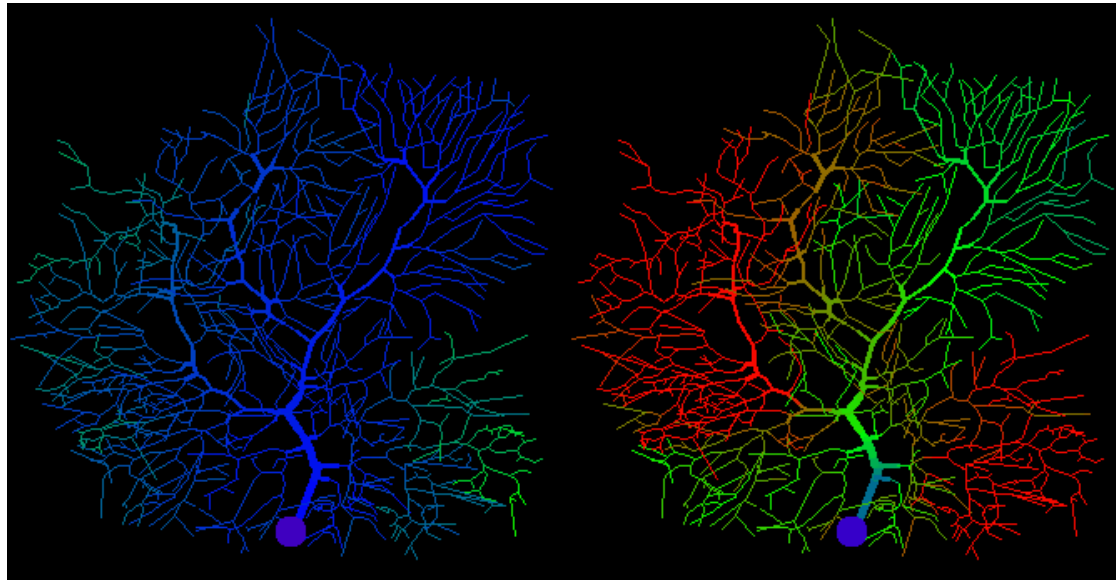
- Cognition and Behaviour
- Distributed systems (e.g. motor system, visual system)
- Brain areas (e.g. cerebellum, hippocampus)
- Local circuits
- Complex neurons
- Dendritic subunits
- Synapses
- Molecules (ion channels, second messengers, DNA)

Methods of brain research

- Psychological experiments
- Behavioural studies (e.g. conditioning, maze learning)
- Non-invasive methods: fMRI, PET, MEG, EEG
- Electrophysiology: extracellular single and multi-unit recordings and patch-clamp recordings from
 - Awake behaving animals
 - Anaesthetised animals
 - Brain slices
 - Isolated neurons
- Voltage sensitive dyes and calcium imaging
- Anatomical and histological studies: immunohistochemistry, in situ hybridization, dye staining, tracers, degeneration studies (examined with light and electron microscopy)
- Pharmacology: selective block of ion channels and receptors
- Molecular biology: KO mice, site-directed mutagenesis
- Computational neuroscience and neural networks: computer simulations and mathematical analyses

Computational model at different levels of complexity

- Detailed models of complex neurons: assume that the neuronal morphology and the behaviour of ion channels are relevant



- Simple models of spiking neurons: assume that the ability to fire spikes is important, study different types of neural coding
- Abstract neural network models: focus on the effect of network connectivity, study general principles of learning and memory

Bibliography

- Purves et al. Neuroscience. Sinauer, 2008.
- Hertz, Krogh & Palmer. Introduction to the theory of neural computation. Addison-Wesley, 1991.
- Gurney. An introduction to neural networks. CRC press, 1997.
- Churchland & Sejnowski. The computational brain. MIT press, 1992.
- Trappenberg. Fundamentals of computational neuroscience. Oxford University Press, 2010.
- Hubel. Eye, Brain and Vision. Scientific American, 1988.
<http://hubel.med.harvard.edu/bcontext.htm>

Summary ?

Questions

- 1) We have discussed the coding of information by biological neurons and networks. Give examples of analog and digital signals in the brain. What are the advantages and disadvantages of analog and digital signalling?
- 2) Why is the formation of ocular dominance columns an example for unsupervised learning? What are the main features of supervised and unsupervised learning?
- 3) Give examples of possible computational functions of lateral inhibition.